

The Southeast Blueprint Changelog

What has changed with each version of the Southeast Blueprint?

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What is a Changelog?

Changelogs are commonly used in software development. According to the [Keep a Changelog project](#), a changelog contains “a curated, chronologically ordered list of notable changes for each version of a project.” Changelogs make it easier for both project users and project contributors to see what notable changes have been made between each version. When something changes, people often want to know both why and how. Here, we’re applying these principles to the Southeast Blueprint.

The Evolution of the Southeast Blueprint

As a living spatial plan, the Southeast Blueprint has evolved over time through an iterative revision process, improving its accuracy, spatial extent, consistency, and utility to conservation professionals. The revision cycle is driven by input from experts and Blueprint users, as well as the availability of better data. This changelog is intended to capture the changes made to the Southeast Blueprint since the first version was released in December 2016.

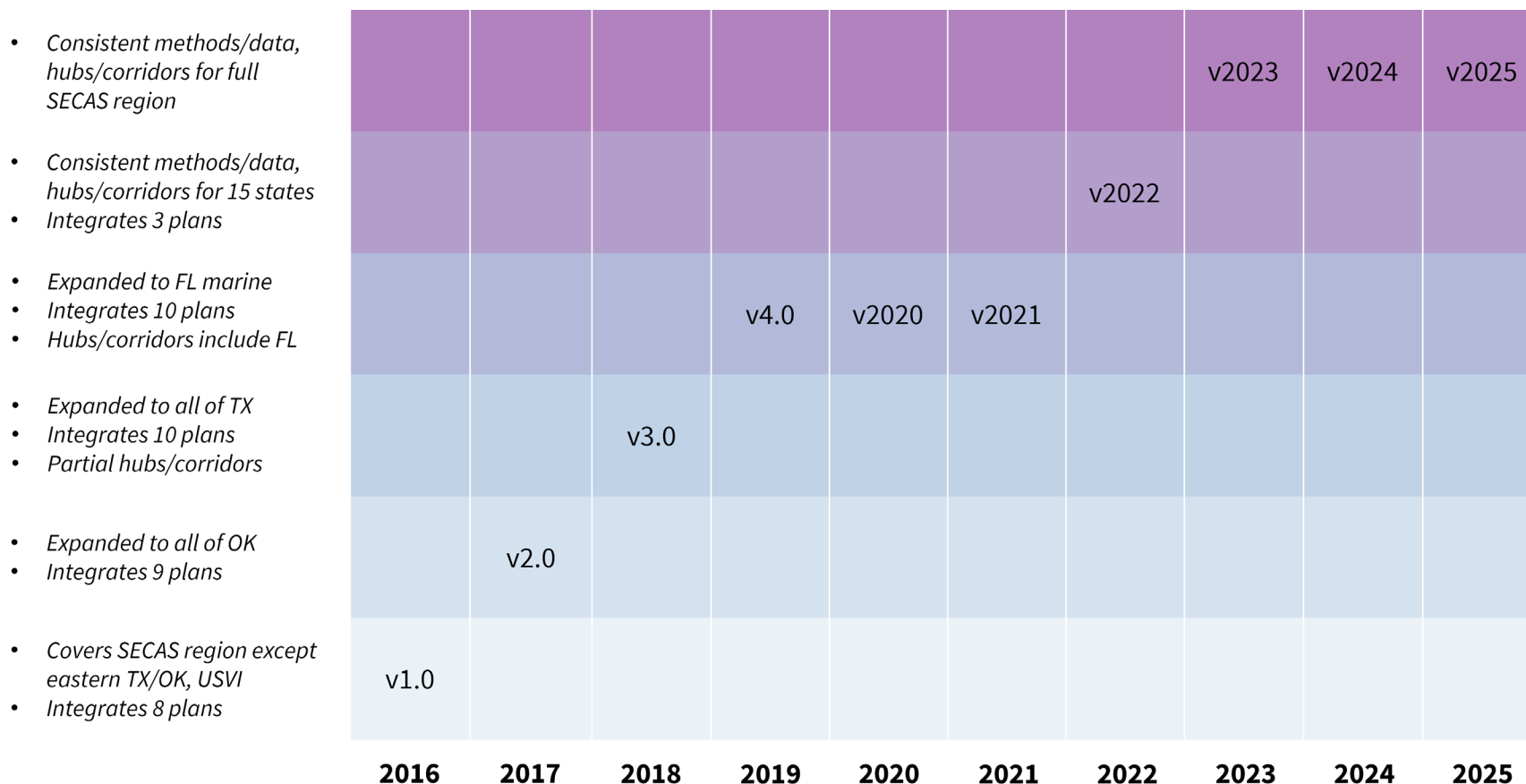


Figure 1. Major changes to the Southeast Blueprint over time by version (where v1.0 represents Southeast Blueprint version 1.0, etc.).

Southeast Blueprint 2025

Version 2024 of the Southeast Blueprint was released in November 2025. This version updated the Blueprint priorities in both the Caribbean and continental Southeast. It used refined prioritization and connectivity methods and incorporated a number of new and improved indicators.

Changes to the Indicators

Summary of Indicator Changes

Table 1. The 68 indicators that drive priority in Southeast Blueprint 2025 and their changes from Southeast Blueprint 2024. Changes are represented by:

o - the indicator was included in a previous Blueprint version and did not change

Δ - the indicator changed from a previous version

- - the indicator has been removed/replaced and is no longer used in this version of the Blueprint

+ - the indicator is new and not included in previous Blueprint versions

Category	Southeast Blueprint 2025 Indicator	Status	Changes from Southeast Blueprint 2024
Terrestrial	Caribbean greenways & trails	o	
	Caribbean habitat patch size (large islands)	o	
	Caribbean habitat patch size (small islands)	o	
	Caribbean island habitat	o	
	Caribbean karst habitat	o	
	Caribbean landscape condition	o	
	Caribbean low-urban historic landscapes	o	
	Caribbean reforestation potential	o	
	Caribbean urban park size	o	
	East Coastal Plain open pine birds	o	
	Fire frequency	o	
	Grassland & savanna extent	Δ	Previously named grasslands and savannas (“extent” added to distinguish from new

Category	Southeast Blueprint 2025 Indicator	Status	Changes from Southeast Blueprint 2024
			restoration indicator); added new known sites; improved methods to capture tree encroachment; decoupled from landscape condition using new approach for clustering based on grassland/savanna “hubs”; fixed misclassification issues in wet longleaf pine and historic shrublands
	Grassland & savanna restoration	+	New indicator added
	Greenways & trails	o	
	Imperiled amphibians & reptiles	+	New amphibian and reptile areas indicator replaced amphibian and reptile areas
	Amphibian & reptile areas	-	
	Imperiled mammals	+	New indicator added
	Intact habitat cores	o	
	Landscape condition	Δ	Updated to capture additional forms of habitat alteration (i.e. reservoirs and tree encroachment in Western historically treeless grasslands); updated to use new grassland/savanna extent indicator; improved accuracy around outer edges of Blueprint extent; refined approach to excluding marine areas from extent
	Mississippi Alluvial Valley forest birds - protection	o	
	Mississippi Alluvial Valley forest birds - reforestation	o	
	Playas	o	
	Potential access to parks	Δ	Updated to newer versions of source data
	Puerto Rico potential access to parks	+	New indicator added
	Resilient terrestrial sites	Δ	Updated to fix minor overlap with resilient coastal sites and better match source data

Category	Southeast Blueprint 2025 Indicator	Status	Changes from Southeast Blueprint 2024
	River cane restoration	+	New indicator added
	South Atlantic forest birds	o	
	South Atlantic low-urban historic landscapes	o	
	Urban park size	o	
	U.S. Virgin Islands imperiled terrestrial species	+	New indicator added
	West Coastal Plain & Ouachitas forested wetland birds	o	
	West Coastal Plain & Ouachitas open pine birds	o	
	West Gulf Coast mottled duck nesting	o	
Freshwater	Atlantic migratory fish habitat	o	
	Caribbean natural landcover in floodplains	o	
	Caribbean network complexity	o	
	Caribbean permeable surface	o	
	Floodplain inundation	+	New indicator added
	Gulf migratory fish connectivity	o	
	Imperiled aquatic species	o	
	Lakes & reservoirs	+	New indicator added
	Natural landcover in floodplains	o	
	Network complexity	o	
	Permeable surface	o	
Coastal & marine	Atlantic coral & hardbottom	o	
	Atlantic deep-sea coral richness	o	
	Atlantic estuarine fish habitat	o	
	Atlantic marine birds	o	
	Atlantic marine mammals	o	

Category	Southeast Blueprint 2025 Indicator	Status	Changes from Southeast Blueprint 2024
Coastal & marine	Caribbean beach habitat	o	
	Caribbean coastal shoreline condition	o	
	Caribbean fish hotspots	o	
	Caribbean fish nursery habitat	o	
	Caribbean mangrove migration	+	New indicator added
	Caribbean seagrass	o	
	Caribbean shallow hardbottom & coral	o	
	Coastal shoreline condition	o	
	Estuarine coastal condition	o	
	Gulf coral & hardbottom	o	
	Gulf deep-sea coral richness	o	
	Gulf marine mammals	o	
	Gulf sea turtles	o	
	Island habitat	o	
	Marine highly migratory fish	o	
	Resilient coastal sites	Δ	Updated to fix minor overlap with resilient terrestrial sites and better match source data; added new “sea level rise areas” class
	Seagrass	o	
	South Atlantic beach birds	o	
	South Atlantic maritime forest	o	
	Stable coastal wetlands	o	

New Indicators Added

Continental

Six new indicators were developed for the terrestrial and aquatic continental area in Blueprint 2025.

Terrestrial

- **Grassland & savanna restoration** was added as a new terrestrial indicator to complement grassland and savanna extent (previously named grasslands and savannas). While the extent indicator focuses on current known, likely, and predicted grasslands, this new indicator focuses on potential restoration opportunities.
- **Imperiled amphibians & reptiles** was added as a new terrestrial indicator, replacing amphibian and reptile areas. This new indicator identifies important areas for priority herpetofauna at a much finer spatial scale than the previous indicator. It is also data-driven, rather than using coarse expert-defined polygons.
- **Imperiled mammals** was added as a new terrestrial indicator, complementing the approach used for imperiled amphibians and reptiles, but focused on priority mammal species.
- **River cane restoration** was added as a new terrestrial indicator to identify places for river cane restoration and co-stewardship near the lands of federally recognized Tribes. River cane is a bamboo species native to the Southeast that has declined dramatically from its historic range but has significant cultural and ecological value.

Freshwater

- **Floodplain inundation** was added as a new freshwater indicator to measure the hydrologic function of the floodplain based on the frequency of flooding. Identifying areas that stay wet or dry too consistently can signal the potential loss of hydrologic function from human alteration (e.g., dams, dikes, levees, and stream incision due to erosion).
- **Lakes & reservoirs** was added as a new freshwater indicator to assess the condition of lakes and reservoirs based on the degree of alteration in the upstream watershed. The addition of this indicator made it possible for the Blueprint to prioritize within the open waters of reservoirs for the first time.

Caribbean

Three new indicators were developed for the terrestrial Caribbean area in 2025.

Terrestrial

- **Puerto Rico potential access to parks** was added as a new terrestrial indicator. This is an analog to the continental potential access to parks indicator and uses the same source data. The data provider expanded the source data to Puerto Rico, but not the U.S. Virgin Islands, and changes to the Zonation methods in Blueprint 2025 made it possible to use this indicator only in Puerto Rico without pulling priority away from the U.S. Virgin Islands.

- **U.S. Virgin Islands imperiled terrestrial species** was added as a new terrestrial indicator. This is an analog to the continental imperiled mammals and amphibian and reptile indicators, but uses different source data. The source data is not available in Puerto Rico, and changes to the Zonation methods in Blueprint 2025 made it possible to use this indicator only in the U.S. Virgin Islands without pulling priority away from Puerto Rico.

Coastal & Marine

- **Caribbean mangrove migration** was added as a new coastal indicator to represent current and potential future mangrove locations.

Old Indicators Removed

Continental

Terrestrial

- **Amphibian and reptile areas** was removed as an indicator, replaced by imperiled amphibians and reptiles (see New Indicators Added section above).

Indicators Changed

Continental

Terrestrial

- **Grasslands & savannas** was renamed **grassland & savanna extent** to distinguish it from the new grassland and savanna restoration indicator. We also updated the indicator in several ways. We added numerous known grassland/savanna sites from a variety of partners and datasets. We fixed a misclassification issue with wet longleaf pine savannas by using LANDFIRE Existing Vegetation Type data to add wet longleaf as possible savanna landcover where NLCD classified the area as forested wetland. We improved our approach to handling tree encroachment in historically treeless grasslands by using LANDFIRE Biophysical Settings (BPS) to identify historic grasslands in the western part of the region that wouldn't typically have trees, and applying stricter canopy cover limits from NLCD canopy cover data. We also used LANDFIRE BPS to fix a misclassification issue where using historic fire return intervals brought in historic shrublands as historic grasslands. In this update, rather than using landscape condition to elevate potential grasslands/savannas within more natural landscapes, we used a new approach to elevate potential grasslands/savannas within grassland/savanna "hubs". These hubs were defined as HUC12 subwatersheds that contain at least one known or likely grassland/savanna patch more >5 acres in size. This change improved clustering of higher indicator values and allowed us to decouple grassland and savanna extent from landscape condition, which streamlined the indicator update process.

Finally, we updated the approach for excluding marine areas from the spatial extent of the indicator for consistency with methods used in other 2025 indicator updates, such as landscape condition and the new imperiled species indicators.

- **Landscape condition** was updated in numerous ways. It captures two new forms of habitat alteration. This version now treats Western historically treeless grasslands experiencing tree encroachment as altered, using LANDFIRE BPS and NLCD tree canopy data. It also treats the open water areas of reservoirs as altered. In addition, it uses the latest update to the grassland and savanna extent indicator to correct for misclassification in grasslands/savannas. We updated the approach for excluding marine areas from the spatial extent of the indicator for consistency with methods used in other 2025 indicator updates, such as grassland and savanna extent and the new imperiled species indicators. Finally, we ran this indicator in a buffer area around the SECAS geography to improve the accuracy of the landscape condition calculations along the outer edges of the Blueprint extent.
- **Potential access to parks** was updated to include the latest source data. It now uses the Trust for Public Land's 2024 ParkServe priority areas and the Centers for Disease Control's 2022 Social Vulnerability Index.
- **Resilient terrestrial sites** was updated from The Nature Conservancy's (TNC) combined terrestrial and coastal resilience data, rather than the terrestrial-only Resilient Land. Using TNC's combined data eliminated the need to remove overlap between the separate terrestrial and coastal layers, which allowed us to fix a raster snapping error where the two resilient sites indicators overlapped in a few pixels. The combined dataset is not any newer than the separate terrestrial one, but better matches how TNC serves up the resilience data in their online viewer.

Coastal & Marine

- **Resilient coastal sites** was updated from The Nature Conservancy's (TNC) combined terrestrial and coastal resilience data, rather than the coastal-only Resilient Coastal Sites. Using TNC's combined data eliminated the need to remove overlap between the separate terrestrial and coastal layers, which allowed us to fix a raster snapping error where the two resilient sites indicators overlapped in a few pixels. The combined dataset is not any newer than the separate coastal one, but better matches how TNC serves up the resilience data in their online viewer. This change also allowed us to add TNC's "sea-level rise (SLR) area" class to the indicator legend, which provides a more complete characterization of coastal resilience. This class denotes locations that are not part of a current tidal complex or future marsh migration space, but are at risk of inundation under 6 feet of SLR. Because they are not current or future marsh, they are outside the scope of the coastal resilience analysis. TNC also does not score them in their terrestrial resilience analysis because they are likely to be inundated in the future. In the previous version, many of these sea-level rise area pixels were excluded from the indicator, while some did receive resilience scores.

Changes to Connectivity

In Blueprint 2025, we updated both the continental and Caribbean connectivity analysis. This is the first Caribbean hubs and corridors update since we expanded the Blueprint to that geography in 2023. Our primary goal in the 2025 update was to improve the alignment of the corridors with other established connectivity efforts in the continental Southeast, and to encourage more corridors to run through inland areas in the Caribbean. We continued to use the same version of Linkage Mapper as in the previous two cycles.

- Southeast Blueprint 2025 priorities were used to define the hubs and resistance raster.
- We updated the protected areas data used in the hubs and corridors to the newest version of PAD-US (version 4.1).

Continental

- We continued to run Linkage Mapper within hexagons to break up the geography into computationally manageable chunks. This year, we returned to the 2023 hexagon size of 500 km². In 2024, we experimented with increasing the hexagon size to 600 km², but this caused issues with file size and model run time. As a result, this year's continental connectivity analysis required 15 separate hexagon runs instead of 12.
- We modified our use of reservoirs in the hubs and corridors. In previous versions of the Blueprint, open water areas of reservoirs were excluded from the Zonation results due to a lack of indicator data to represent their condition; for consistency, we removed reservoirs from the continental hubs and corridors as well. This year, with the creation of the new lakes and reservoirs indicator, open water parts of reservoirs became eligible to prioritize in Zonation (and, by extension, the Blueprint) for the first time. As a result, we stopped removing reservoirs from the corridors entirely. In addition, rather than removing reservoirs from the final hubs, we only removed them from the protected areas component of the potential hubs. Reservoirs were still eligible to become hubs if they were included in the top 10% of the Zonation results, but not just because of their protected status, given that they are innately altered systems.
- We reintroduced methods from 2023 that constrained the marine corridors by marine mammal and turtle movement areas to encourage them to follow routes important for these species. We dropped these in 2024 as part of an effort to simplify and streamline the corridor methods, but found that they helped refine the marine corridor routing.
- This year, we wanted to encourage continental corridors to route through other established connectivity priorities identified by partners, particularly state wildlife agencies. To accomplish this, we pulled in connectivity priorities for five states: the VA Wildlife Corridor Action Plan, the FL Ecological Greenways Network/Florida Wildlife Corridor, and connectivity-based Conservation Opportunity Areas from the State Wildlife Action Plans for GA, SC, and NC. We made these connectivity priorities easier to for corridors move through by lowering their

value in the resistance raster by up to 5. We hope to expand this approach to other parts of the SECAS geography as more state and local corridor plans become available.

- We also modified our hub methodology to reduce the influence of extremely long, linear protected areas, such as the Natchez Trace Parkway in MS. We accomplished this by shrinking and expanding the protected areas layer by three pixels to essentially remove protected areas that are three or fewer pixels wide. While parkways are an important cultural and recreational resource, and can still influence corridor routing, they do not align with the intent of the hubs as large contiguous patches of conservation priorities and/or protected areas.

Caribbean

- The 2025 Blueprint represents the first update to the Caribbean hubs and corridors since the Blueprint was first expanded to the Caribbean in 2023. In 2024, we did not update the Blueprint in the Caribbean, so there was no new connectivity analysis for the Caribbean.
- We decreased the size of hubs in the Caribbean geography from 200 ha/~500 ac to 50 ha/~125 ac. This created more inland hubs in the islands' interiors and encouraged corridors to route across the islands, instead of around them through the marine environment. Creating more inland corridors in the U.S. Caribbean, particularly on St. Croix, was a priority for the 2025 Blueprint update based on partner feedback. Compared to the continental Southeast, decreasing hub size in the U.S. Caribbean is much more computationally feasible given the relatively small size of the geography.

Changes to the Zonation Results

Continental

Improvements to the Overall Zonation Modeling Approach

Blueprint 2025 incorporated a key Zonation methods change in the continental geography:

- We stopped using a reservoirs mask to remove the open water area of reservoirs from the continental Zonation results, and by extension, the Blueprint priorities. The introduction of the new lakes and reservoirs indicator allowed us to incorporate reservoir condition into the priorities and represent them appropriately within the Blueprint for the first time. As a result, it was no longer necessary to rebalance the Zonation results to account for the reservoir area removed from each zone when calculating the final Blueprint priority classes.

Weighting Rules

Inland

- We modified most of the rules for weighting indicators in the inland continental area. The only two rules we left unchanged applied to subregional indicators that don't cover most of their target ecosystem and indicators with limited overlap of below-average values in a given zone.
 - We expanded the scope of the weighting rule that applies to indicators with a coarse spatial scale to also include indicators that use coarse information. Coarse information refers to broad definitions of potential species habitat and species ranges, broad definitions of habitat condition, and simplified site-specific variation in ecosystem health and habitat value. To achieve this, we reduced the weight of the estuarine coastal condition, imperiled mammals, imperiled amphibians and reptiles, grassland and savanna extent, and grassland and savanna restoration indicators by 0.7 across all zones. This rule previously applied only to estuarine coastal condition and the old amphibian and reptile areas indicator, and applied a weight of 0.5 across all zones.
 - We adjusted the thresholds used in the spatially limited indicator rule. For indicators where the proportion of ≥ 1 values was between 0.09 and 0.5, we set their weights equal to that proportion (the previous version applied this adjustment for indicators where the proportion of ≥ 1 values was between 0.1 and 0.5). For indicators where the proportion of ≥ 1 values was ≤ 0.09 (previously ≤ 0.1), we set their weights equal to that proportion multiplied by 3.
- In addition to modifying the rules themselves, we also revised some of the exceptions to the standard rules.
 - We adjusted the previous exception to the spatially limited indicators rule that applied to landscape condition and intact habitat cores in the Arid West. This year, we applied the exception only to intact habitat cores in the Arid West, giving it the same weight of 0.7 as in the previous version, while landscape condition received the standard weighting rule. The weight reduction was no longer needed now that landscape condition included impacts of tree encroachment, a major impact to conditions in the Arid West.
 - We added a new exception to the spatially limited indicators rule specific to lakes and reservoirs. The open water areas of lakes and reservoirs are difficult to prioritize because none of the indicators provide highly detailed information about the relative importance of pixels in these areas. This can result in computational challenges for Zonation, particularly when the weight for this indicator is too high. To address this, we followed the rule by setting the lakes and reservoirs weight equal to the proportion of ≥ 1 values, but we did not multiply that proportion by 3 when the proportion was ≤ 0.09 . This kept the weight low enough to avoid computational issues in Zonation.

- We adjusted the exception to the spatially limited indicators rule for indicators right on the edge of weighting thresholds where applying the standard weight causes issues. We continued the previous year's exception for intact habitat cores in the Coastal Zone, but instead of doubling the weight to bring it closer to 1, we simply set it to 1. We also removed the exception for subregional inputs in the Central West Zone, as it was no longer necessary to balance priorities across subregions in the zone.

Caribbean

Improvements to the Overall Zonation Modeling Approach

- In 2025, we split the previous single Caribbean Zone into two separate zones: one for Puerto Rico and one for the U.S. Virgin Islands. As these sections of the Southeast Blueprint are non-contiguous, they were easy to separate. We made this change largely because we introduced two new indicators that did not have data available for the entire Caribbean: Puerto Rico potential access to parks and U.S. Virgin Islands imperiled terrestrial species. Running Puerto Rico and the USVI in separate Zonation zones allowed us to avoid penalizing (i.e. pulling Blueprint priority away from) the part of the Caribbean geography not covered by each indicator. This change also lays the groundwork for ensuring the Blueprint better reflects differences not only in data availability, but island size, biodiversity, other aspects of the two territories. For example, because Puerto Rico is a larger land mass, it has larger maximum patch sizes compared to the USVI. It also has more extensive freshwater aquatic resources stemming from more permanent waterbodies and waterways.

Weighting Rules

- The Caribbean portion of the SECAS geography only had one previous weighting rule for spatially limited indicators, and it did not change from 2024. However, we did add a new exception to that weighting rule:
 - In 2025, two spatially limited indicators on the edge of the standard weighting rule threshold required special exceptions to solve problems in Zonation. In the Puerto Rico Zone, Caribbean habitat patch size (large Islands) was just over the 0.5 weighting threshold. It was being overprioritized, so we kept the weight equal to the proportion of ≥ 1 values and did not increase the weight to 1. The U.S. Virgin Islands imperiled terrestrial species indicator was being underprioritized and was close to a 0.3 threshold we used in previous versions of the Blueprint, so we doubled its weight in the U.S. Virgin Islands Zone to compensate.

Southeast Blueprint 2024

Version 2024 of the Southeast Blueprint was released in October 2024. This version focused on updating the priorities in the continental Southeast. It used refined prioritization and connectivity methods and incorporated a number of new and improved indicators for both inland and marine areas. The U.S. Caribbean priorities were not updated at all in version 2024.

Changes to the Indicators

Summary of Indicator Changes

Table 2. The 60 indicators that drive priority in Southeast Blueprint 2024 and their changes from Southeast Blueprint 2023. Changes are represented by:

o - the indicator was included in a previous Blueprint version and did not change

Δ - the indicator changed from a previous version

- - the indicator has been removed/replaced and is no longer used in this version of the Blueprint

+ - the indicator is new and not included in previous Blueprint versions

Category	Southeast Blueprint 2024 Indicator	Status	Changes from Southeast Blueprint 2023
Terrestrial	Amphibian & reptile areas	+	New amphibian and reptile areas indicator replaced South Atlantic amphibian and reptile areas
	South Atlantic amphibian & reptile areas	-	
	Caribbean greenways & trails	o	
	Caribbean habitat patch size (large islands)	o	
	Caribbean habitat patch size (small islands)	o	
	Caribbean island habitat	o	
	Caribbean karst habitat	o	
	Caribbean landscape condition	o	
	Caribbean low-urban historic landscapes	o	
	Caribbean reforestation potential	o	
	Caribbean urban park size	o	

Category	Southeast Blueprint 2024 Indicator	Status	Changes from Southeast Blueprint 2023
Terrestrial	East Coastal Plain open pine birds	o	
	Fire frequency	o	
	Grasslands & savannas	+	New grasslands and savannas indicator replaced Interior Southeast grasslands and Great Plains perennial grasslands
	Great Plains perennial grasslands	-	
	Interior Southeast grasslands	-	
	Greenways & trails	o	
	Intact habitat cores	Δ	Updated to newer version of source data
	Landscape condition	+	New indicator added
	Mississippi Alluvial Valley forest birds - protection	o	
	Mississippi Alluvial Valley forest birds - reforestation	o	
	Potential access to parks	Δ	Previously named equitable access to potential parks (shortened for clarity)
	Playas	o	
	Resilient terrestrial sites	o	
	South Atlantic forest birds	o	
	South Atlantic low-urban historic landscapes	o	
	Urban park size	Δ	Updated to newer versions of source data; added beaches from OpenStreetMap and missing U.S. FWS protected areas; extended to nearshore marine parks; added new class for parks <5 acres
	West Coastal Plain & Ouachitas forested wetland birds	o	
	West Coastal Plain & Ouachitas open pine birds	o	
	West Gulf Coast mottled duck nesting	o	

Category	Southeast Blueprint 2024 Indicator	Status	Changes from Southeast Blueprint 2023
Freshwater	Atlantic migratory fish habitat	o	
	Caribbean natural landcover in floodplains	o	
	Caribbean network complexity	o	
	Caribbean permeable surface	o	
	Gulf migratory fish connectivity	o	
	Imperiled aquatic species	Δ	Updated to newer version of source data; switched from counting number of SGCN to Regional SSGN; expanded to WV to eliminate the need for a separate indicator
	West Virginia imperiled aquatic species	-	
	Natural landcover in floodplains	Δ	Updated to newer version of source data; fixed missing catchments error
	Network complexity	Δ	Fixed missing catchments error
	Permeable surface	Δ	Updated to newer version of source data; fixed missing catchments error
Coastal & marine	Atlantic coral & hardbottom	Δ	Added new and improved data sources; removed lower probability of hardbottom areas; improved data ranking; redefined Atlantic extent to better align with Gulf version; redefined inland extent to eliminate large buffers
	Atlantic deep-sea coral richness	o	
	Atlantic estuarine fish habitat	o	
	Atlantic marine birds	o	
	Atlantic marine mammals	o	
	Caribbean beach habitat	o	
	Caribbean coastal shoreline condition	o	
	Caribbean fish hotspots	o	
	Caribbean fish nursery habitat	o	
	Caribbean seagrass	o	

Category	Southeast Blueprint 2024 Indicator	Status	Changes from Southeast Blueprint 2023
Coastal & marine	Caribbean shallow hardbottom & coral	o	
	Coastal shoreline condition	o	
	Estuarine coastal condition	o	
	Gulf coral & hardbottom	Δ	Added new and improved data sources; removed unreliable decommissioned oil and gas platform data (rigs-to-reef sites still captured as artificial reefs); improved data ranking; redefined Gulf extent to better align with Atlantic equivalent; redefined inland extent to eliminate large buffers
	Gulf deep-sea coral richness	o	
	Gulf marine mammals	o	
	Gulf sea turtles	Δ	Adjusted lowest percentile threshold to eliminate artifact
	Island habitat	o	
	Marine highly migratory fish	o	
	Resilient coastal sites	o	
	Seagrass	o	
	South Atlantic beach birds	o	
	South Atlantic maritime forest	o	
	Stable coastal wetlands	o	

New Indicators Added

Continental

Many new indicators were developed for the marine continental area as part of expanding the Blueprint to the rest of the offshore waters of the Atlantic and Gulf.

Terrestrial

- **Amphibian and reptile areas** was added as a new terrestrial indicator, replacing South Atlantic amphibian and reptile areas. This new indicator draws on the same approach based on Priority Amphibian and Reptile Conservation Areas (PARCAs), but expands its spatial extent to all portions of the SECAS geography where PARCAs have been identified (all SECAS states except Virginia and Kentucky).
- **Grasslands and savannas** was added as a new terrestrial indicator, replacing both Interior Southeast grasslands and Great Plains perennial grasslands. Grasslands and savannas uses improved methods and data to better represent known, predicted, and historic grasslands across the entire Southeast.
- **Landscape condition** was added as a new terrestrial indicator, mirroring the approach used for Caribbean landscape condition. The continental version adds an additional “very natural” class to reflect the relative maximum size potential compared to the Caribbean. It also uses additional methods and data to distinguish naturally barren areas from altered barren areas like quarries, and to reflect natural vs. altered forest composition.

Old Indicators Removed

Continental

Terrestrial

- **South Atlantic amphibian and reptile areas** was removed as an indicator, replaced by amphibian and reptile areas (see New Indicators Added section above).
- **Interior Southeast grasslands** was removed as an indicator, replaced by grasslands and savannas (see New Indicators Added section above).
- **Great Plains perennial grasslands** was removed as an indicator, replaced by grasslands and savannas (see New Indicators Added section above).

Freshwater

- **West Virginia imperiled aquatic species** was removed as an indicator because the full imperiled aquatic species indicator was expanded to cover West Virginia. The Southeast Aquatic Resources Partnership (SARP) recently expanded their geography to include West Virginia and began collecting data for that state. Compared to the previous indicator, SARP's approach offers much more recent data and a consistent approach to the rest of the Southeast.

Indicators Changed

Continental

Terrestrial

- **Intact habitat cores** was updated to incorporate the latest version of the source data from Esri, which is now based on the 2019 National Land Cover Database (NLCD) instead of 2011. We also updated the version of the NLCD used to remove deep marine areas from the zero class of the indicator from 2019 to 2021.
- **Urban park size** was updated to incorporate newer source data and improve consistency with the Caribbean version of the indicator. We updated it to version 3.0 of the Protected Areas Database of the United States (PAD-US), 2020 Census Urban Areas, and impervious surface from the 2021 NLCD. We added an authoritative U.S. Fish and Wildlife Service protected areas dataset to fill gaps in PAD-US. We included nearshore marine protected areas. We also included beaches from OpenStreetMap. Finally, we added a separate class for parks less than 5 acres in size, rather than lumping them into the "not identified as an urban park" class.
- **Equitable access to potential parks** was renamed **potential access to parks** to shorten and clarify the name.

Freshwater

- **Imperiled aquatic species** was updated to a fresh download of the source data from the Southeast Aquatic Resources Partnership (SARP). SARP recently expanded their geography to include West Virginia and began collecting data for that state, so the indicator was expanded to cover West Virginia. We also switched from using the number of Species of Greatest Conservation Need (SGCN) observed in a watershed to the number of Regional SGCN. This improved regional consistency by better accounting for variation in the number of SGCN each state identifies.
- **Natural landcover in floodplains** was updated to the 2021 NLCD. To correct a previous known issue related to selecting a subset of catchments, it was also expanded to all medium resolution catchments within the SECAS geography, which filled a gap in Lake Pontchartrain.

- **Network complexity** was updated to correct a previous known issue related to selecting a subset of catchments. It was expanded to all medium resolution catchments within the SECAS geography, which filled a gap in Lake Pontchartrain.
- **Permeable surface** was updated to the 2021 NLCD. To correct a previous known issue related to selecting a subset of catchments, it was also expanded to all medium resolution catchments within the SECAS geography, which filled a gap in Lake Pontchartrain.

Coastal & Marine

- **Atlantic coral and hardbottom** was updated in numerous ways. We modified the extent to eliminate the large inland buffers previously included in the lowest “not identified as hardbottom” class by using elevation data to constrain the extent rather than distance onshore. We also eliminated the previous overlap with the Gulf coral and hardbottom indicator, which will make it easier to analyze the two indicators for Blueprint user support. The new Atlantic/Gulf split was defined using the boundaries of available datasets, the Blueprint extent and subregions, marine ecoregions, and elevation. In the input data and ranking, in the top class, we used a fresh download of points from the NOAA Deep-Sea Coral and Sponge Map Portal and included all coral and sponge points rather than targeting particular coral taxa. We decoupled natural from artificial hardbottom so that natural hardbottom habitat scored more highly, correcting an oversight from the previous version. We added new data representing coral and hardbottom locations in nearshore waters off the Atlantic coast of Florida. We split artificial reefs and shipwrecks into different classes to score artificial reefs more highly, reflecting their intentional siting for conservation benefit. In addition to NOAA’s regional artificial reef data, we added more comprehensive state data on Florida artificial reef locations (the NOAA dataset matched state sources for all other Atlantic states). Finally, we excluded the lowest two probability quantiles from NOAA’s predictive hardbottom models for more consistency with the Gulf version of the indicator, which does not attempt to provide a continuous probability surface and only highlights likely and known hardbottom features.
- **Gulf coral and hardbottom** was updated in numerous ways. We modified the extent to eliminate the large inland buffers previously included in the lowest “not identified as hardbottom” class by using elevation data to constrain the extent rather than distance onshore. We also eliminated the previous overlap with the Atlantic coral and hardbottom indicator, which will make it easier to analyze the two indicators for Blueprint user support. The new Gulf/Atlantic split was defined using the boundaries of available datasets, the Blueprint extent and subregions, marine ecoregions, and elevation. In the input data and ranking, in the top class of the indicator, we used a fresh download of points from the NOAA Deep-Sea Coral and Sponge Map Portal and included all coral and sponge points rather than targeting particular coral taxa. We added new data from Florida representing coral and hardbottom locations in nearshore waters of the Gulf. We also incorporated known hardbottom areas from The Nature Conservancy’s South Atlantic Bight Marine Assessment

that were already used in the Atlantic version but also spilled over slightly into the Gulf. The additional known hardbottom data required a new natural hardbottom class. Because these updates did a better job of highlighting the most important areas within Coral Habitat Areas of Particular Concern (HAPCs), we removed the coarse HAPCs as an input. We split artificial reefs and shipwrecks into different classes to score artificial reefs more highly, reflecting their intentional siting for conservation benefit. We used a fresh download of shipwrecks data from a different NOAA source that provided confidence information for wreck locations, and used only high and medium confidence points. We removed the decommissioned oil and gas platforms from the previous human-created hardbottom class to address a known issue with the reliability of that dataset, and because platforms repurposed through “rigs to reefs” programs were captured by state artificial reef locations. We replaced the NOAA regional artificial reef dataset with more comprehensive state data on Gulf artificial reef locations.

- **Gulf sea turtles** was modified slightly to eliminate a minor data artifact. The bottom class of the indicator was adjusted to include only the 65th-70th percentile of important areas for sea turtles, rather than the 60-70th percentile.

Changes to Connectivity

Continental

We continued to use the same version of Linkage Mapper for our connectivity analysis in Southeast Blueprint 2024. Our primary goal in the 2024 update was to simplify the complex methods for the continental geography as a foundation for further refinement. We made several changes to the continental connectivity analysis:

- Southeast Blueprint 2024 priorities were used to define the hubs and resistance raster.
- We ran inland and marine continental corridors together using the same methods, rather than splitting them out with separate methods. As a result, we modified the hexagon placement for each Linkage Mapper run to encompass the combined continental area. This year, we experimented with increasing the hexagon size from 500 km² to 600 km², though we will likely go back to the smaller hexagons in the future due to issues with file size.
- We no longer used large estuaries as hubs in the marine environment. Instead, we matched the previous inland methods, using 2,000 ha patches of highest priority areas in the Blueprint and/or protected areas. This year was the first time we used marine protected areas as hubs.
- We updated the protected areas data to use the newest version of PAD-US (version 4.0). To correct a documented issue with the omission of Oklahoma Wildlife Management Areas in PAD-US 4.0, we added those back in from PAD-US 3.0 using the same approach as the grasslands and savannas indicator.
- To simplify the methods for 2024, we no longer constrained the marine corridors by marine mammal movement zones, though we plan to investigate further ways to refine marine corridor routing in the future.

- We continued to make urban areas harder to move through by assigning them a higher resistance value. This year, we updated to the medium and high intensity developed classes of the 2021 NLCD.
- To encourage corridors to route through protected areas, even those too small to meet the threshold for a hub, we lowered the resistance value of protected areas by up to 10.
- We continued to align the Southeast Blueprint connectivity analysis with corridors outside our boundaries. Now that the Midwest Landscape Initiative's Blueprint is available, this year we incorporated the hubs and combined Zonation results from Midwest Blueprint 2023 into our hubs and resistance raster wherever they occur outside the Southeast Blueprint extent. In areas not covered by either Blueprint, we continued to use The Nature Conservancy's local connectedness layer from their Resilient Land analysis.

Changes to the Zonation Results

Continental

Improvements to the Overall Zonation Modeling Approach

Blueprint 2024 incorporated several Zonation methods improvements in the continental geography:

- We switched from using Zonation 4 to Zonation 5 in the inland continental area. Since Blueprint 2023 already used Zonation 5 in the marine continental and Caribbean, this was the last portion of the SECAS geography to make the jump to Zonation 5. Zonation 5 offers different settings and rules compared to Zonation 4, but we used the setting most consistent with the core-area Zonation algorithm used previously. The most notable difference is that edge removal is no longer available in Zonation 5, which affects the aggregation of priorities in the final Blueprint. Zonation 5 also no longer requires specifying a warp factor. In addition, it runs much faster and is computationally capable of running larger areas.
- To take advantage of Zonation 5's faster run time and added computational capacity, we no longer ran Zonation within the individual inland continental subregions, but within four larger zones. As a result, each inland subregion no longer has exactly the same proportion of priority in Blueprint 2024. The zones were created by aggregating the previous smaller subregions.
- We continued to run the marine continental area together in one zone, as in the previous version, but modified the subregions within it to eliminate the need to hand-draw a boundary to split the Atlantic and Gulf. Instead of just two Atlantic and Gulf subregions, the Marine Continental Zone now has three subregions: Gulf, Atlantic, and South Florida Marine.

- To better balance priorities across subregions within each zone, we included the subregions as inputs to each Zonation run, as if they were another indicator. This encouraged Zonation to spread out the priorities among different subregions. These methods struck a compromise between having equal proportions of priority in each subregion, as we did by running Zonation 4 within subregions, and allowing the indicators to entirely drive priority within a zone with no attempt to balance across subregions. This hybrid approach was selected and refined based on feedback from a series of focus groups in March 2024.
- Unlike Zonation 4, where we used the edge removal setting, Zonation 5 does not require edges to work from as it prioritizes. This meant that we no longer needed to exclude urban areas from Zonation in the inland continental area this year. While we continued to mask out reservoirs because our current suite of indicators does not do a good job of differentiating the condition of open water areas of reservoirs, urban areas became eligible to be prioritized in the Blueprint based on their indicator scores.
- Because Zonation 4 offers edge removal and Zonation 5 does not, Zonation 4 produced a more aggregated pattern with larger contiguous blocks of priorities, compared to the default settings in Zonation 5, which trend toward a more speckled appearance driven by fine-scale variation in indicator scores. In order to more closely approximate the previous pattern, we used new aggregation methods in Zonation 5 called a “connectivity transformation.” We had Zonation include both the original version of the indicators and connectivity transformed versions of those indicators. This approach is described in the Zonation manual as a potential way for increasing connectivity. This differs from our approach to running Zonation 5 in the marine continental and Caribbean portions of Blueprint 2023, where we did not use additional aggregation.

Weighting Rules

Inland

- We modified the rules for weighting indicators in the inland continental area. The only rule we left unchanged was the one designed to ensure spatially limited indicators were not overprioritized.
 - We removed the previous rule designed to limit the influence of significantly outdated indicator data. This was intended to solve a problem with South Atlantic forest birds, but this year, we addressed it with a more generalized approach that limited the influence of all indicators that do not cover their entire target ecosystem.
 - We added a new rule to limit the influence of coarse-scale indicators. To achieve this, we reduced the weight of the estuarine coastal condition and amphibian and reptile areas indicators by 0.5 across all zones.

- We added a new rule to reduce overprioritization of subregional indicators that don't cover most of their target ecosystem. This applied to nine indicators that only cover specific subregions (e.g., South Atlantic maritime forest only covers a subset of the maritime forest in the Southeast). We reduced the weight of these indicators by 0.5 across all zones.
- We added a new rule to reduce overprioritization of indicators with limited overlap of below-average values in a given zone. This issue occurs in four cases (e.g., low values in imperiled aquatic species spill over slightly into the Marine Zone). To address this, we multiplied the weight of the indicators in the affected zones by 0.1.
- In addition to modifying the rules themselves, we also had to revise the exceptions to the standard rules. We removed all previous exceptions and replaced them with the following:
 - We modified the use of the spatially limited indicators rule for the intact habitat cores and landscape condition indicators in the Arid West Zone, where they do not perform as well. The standard rule caused overprioritization in the Arid West. To address this, we set the weights for these indicators to 0.7 in this zone only.
 - In a few cases, Zonation inputs in certain zones fell right on the edge of a weighting threshold in the spatially limited indicators rule, and we had to create exceptions. We modified the use of the spatially limited indicators rule for intact habitat cores in the Coastal Zone by doubling its standard weight to address underrepresentation. We also did not apply the extra upward adjustment for subregional inputs in the Central West Zone where the proportion of ≥ 1 values was ≤ 0.1 . That would have tripled their weight and overprioritized those subregions.
 - Because of the changes made to inland continental indicators in 2024, as well as the switch to use zones instead of subregions, many of the indicator weights themselves changed. However, we did not document those here because the weights are highly variable from year to year. The changelog only captures changes to the rules that are used to calculate those weights.

Weighting Rules

Marine

- The weighting rules for the marine continental area stayed almost the same, with one minor change. We made a slight adjustment to the rule intended to balance Atlantic and Gulf priorities. Like last year, we again increased the weight of two indicators that were getting significantly underrepresented in the Blueprint priorities—Atlantic coral and hardbottom and Gulf deep-sea coral richness. But instead of increasing them to 0.75, we just raised them to 0.6.
- Because of the changes made to marine continental indicators in 2024, many of the indicator weights themselves changed. However, we did not document those here because the weights are highly variable from year to year. The changelog only captures changes to the rules that are used to calculate those weights.

Southeast Blueprint 2023

Version 2023 of the Southeast Blueprint was released in October 2023. This version expanded consistent methods and indicators to Puerto Rico, the U.S. Virgin Islands, and the surrounding nearshore waters of the U.S. Caribbean, as well as the full U.S. offshore waters of the Atlantic Ocean and Gulf of America within the SECAS geography. For the first time, the Blueprint no longer needed to stitch together any distinct subregional plans.

As a result, the Florida Marine Blueprint and Caribbean Landscape Conservation Design were no longer used as inputs, and it was no longer necessary to provide an “input areas” layer to depict where different inputs were used. The rest of the Blueprint 2023 changelog chapter will focus on changes between Base Blueprint 2022, which was used as the main input to Southeast Blueprint 2022 across 15 states and a portion of the Atlantic marine environment, and Southeast Blueprint 2023.

Changes to the Indicators

Improvements to the Overall Indicator Approach

- Many new indicators were added to cover Puerto Rico, the U.S. Virgin Islands, and parts of the U.S. Caribbean marine environment. Where relevant, we developed Caribbean equivalents for as many of the continental indicators as possible, though the Caribbean analogs often use different data sources due to differences in data availability, as well as lower thresholds for size or length to reflect the smaller size of the Caribbean islands. We also developed several unique indicators specific to the ecosystems and resources of the U.S. Caribbean.
- Many new indicators were added to cover the full extent of the U.S. offshore marine environment in the Atlantic Ocean and Gulf of America within the SECAS geography. Past indicators that previously only extended to the portion of the Atlantic marine environment covered by Base Blueprint 2022 were expanded to the full Atlantic.
- Indicators in Base Blueprint 2022 that were not used across their full extent (due to an oversight in which indicators were included in each subregional Zonation run) were used across their full extent and included in all subregional Zonation runs where they occurred.
- We added meaningful 0 values to as many of the 2023 indicators as possible to better represent the extent of the source data and make them perform better in online tools. This allows users to better distinguish between places not evaluated by an indicator at all (which receive a value of NoData), and places where an indicator was assessed but found not to occur there (which receive a value of 0).

Summary of Indicator Changes

Table 3. The 61 indicators that drive priority in Southeast Blueprint 2023 and their changes from Base Blueprint 2022. Changes are represented by:

o - the indicator was included in a previous Blueprint version and did not change

Δ - the indicator changed from a previous version

- - the indicator has been removed/replaced and is no longer used in this version of the Blueprint

+ - the indicator is new and not included in previous Blueprint versions

Category	Southeast Blueprint 2023 Indicator	Status	Changes from Base Blueprint 2022
Terrestrial	Caribbean greenways & trails	+	New indicator added
	Caribbean habitat patch size (large islands)	+	New indicator added
	Caribbean habitat patch size (small islands)	+	New indicator added
	Caribbean island habitat	+	New indicator added
	Caribbean karst habitat	+	New indicator added
	Caribbean landscape condition	+	New indicator added
	Caribbean low-urban historic landscapes	+	New indicator added
	Caribbean reforestation potential	+	New indicator added
	Caribbean urban park size	+	New indicator added
	East Coastal Plain open pine birds	Δ	Added 0 class to define analysis extent; used across full indicator extent
	Equitable access to potential parks	Δ	Added 0 class
	Fire frequency	Δ	Removed marine areas from 0 class
	Great Plains perennial grasslands	o	No change
	Greenways & trails	Δ	Updated to newer version of source data; improved method of distinguishing trails from sidewalks; added 0 class
	Intact habitat cores	Δ	Removed marine areas from 0 class
	Interior Southeast grasslands	o	No change

Category	Southeast Blueprint 2023 Indicator	Status	Changes from Base Blueprint 2022
Terrestrial	Mississippi Alluvial Valley forest birds - protection	Δ	Added 0 class to define analysis extent; used across full indicator extent
	Mississippi Alluvial Valley forest birds - reforestation	Δ	Added 0 class to define analysis extent; used across full indicator extent
	Playas	Δ	Added 0 class to define analysis extent
	Resilient terrestrial sites	o	No change
	South Atlantic amphibian & reptile areas	Δ	Removed marine areas from 0 class; used full PARCA polygons instead of clipping to South Atlantic LCC boundary; used across full indicator extent
	South Atlantic forest birds	Δ	Used across full indicator extent
	South Atlantic low-urban historic landscapes	Δ	Used across full indicator extent
	Urban park size	Δ	Added 0 class to define analysis extent; buffered small parks; improved methods to capture large contiguous park complexes
	West Coastal Plain & Ouachitas forested wetland birds	Δ	Used across full indicator extent; added 0 class to define analysis extent
	West Coastal Plain & Ouachitas open pine birds	Δ	Used across full indicator extent; added 0 class to define analysis extent
	West Gulf Coast mottled duck nesting	Δ	Used across full indicator extent; added 0 class to define analysis extent
Freshwater	Atlantic migratory fish habitat	Δ	Used across full indicator extent
	Caribbean natural landcover in floodplains	+	New indicator added
	Caribbean network complexity	+	New indicator added
	Caribbean permeable surface	+	New indicator added
	Gulf migratory fish connectivity	Δ	Added 0 class to define analysis extent; used across full indicator extent

Category	Southeast Blueprint 2023 Indicator	Status	Changes from Base Blueprint 2022
Freshwater	Imperiled aquatic species	Δ	Updated to newer version of source data; modified 0 class to define analysis extent and renumbered other classes
	Natural landcover in floodplains	Δ	Added 0 class to define analysis extent
	Network complexity	Δ	Added 0 class to define analysis extent; updated to newer version of source data; expanded to all of WV; filled gaps in catchments with braided streams/loops; switched from medium to high resolution catchments to better align with flowlines
	Permeable surface	o	No change
	West Virginia imperiled aquatic species	Δ	Modified 0 class to define analysis extent and renumbered other classes; changed top class from 5 species to 4+
Coastal & marine	Atlantic coral and hardbottom	+	New Atlantic coral and hardbottom indicator replaced South Atlantic hardbottom and deep-sea coral
	South Atlantic hardbottom & deep-sea coral	-	
	Atlantic deep-sea coral richness	+	New indicator added
	Atlantic estuarine fish habitat	Δ	Used across full indicator extent; added data for FL
	Atlantic marine birds	+	New Atlantic-wide marine birds indicator replaced South Atlantic marine birds
	South Atlantic marine birds	-	
	Atlantic marine mammals	+	New Atlantic-wide marine mammals indicator replaced South Atlantic marine mammals
	South Atlantic marine mammals	-	
	Caribbean beach habitat	+	New indicator added
	Caribbean coastal shoreline condition	+	New indicator added
	Caribbean fish hotspots	+	New indicator added
	Caribbean fish nursery habitat	+	New indicator added
	Caribbean seagrass	+	New indicator added

Category	Southeast Blueprint 2023 Indicator	Status	Changes from Base Blueprint 2022
Coastal & marine	Caribbean shallow hardbottom & coral	+	New indicator added
	Coastal shoreline condition	Δ	Used across full indicator extent
	Estuarine coastal condition	Δ	Added 0 class to define analysis extent; extended further offshore into marine areas
	Gulf coral & hardbottom	+	New indicator added
	Gulf deep-sea coral richness	+	New indicator added
	Gulf marine mammals	+	New indicator added
	Gulf sea turtles	+	New indicator added
	Island habitat	Δ	Previously named islands (added habitat for clarity and consistency with Caribbean version); used across full indicator extent
	Marine highly migratory fish	+	New indicator added
	Resilient coastal sites	Δ	Used across full indicator extent
	Seagrass	+	Previously named seagrasses (made singular for consistency with Caribbean version); updated to newer version of source data
	South Atlantic beach birds	Δ	Improved spatial resolution; removed areas of poor data coverage in VA and the Gulf coast; used across full indicator extent
	South Atlantic maritime forest	Δ	Added 0 class to define analysis extent; used across full data extent; clipped to subregions rather than South Atlantic LCC boundary
	Stable coastal wetlands	Δ	Added 0 class to define analysis extent; modified 1 class to cover other coastal wetlands not identified as stable and renumbered top class to 2

New Indicators Added

Continental

Many new indicators were developed for the marine continental area as part of expanding the Blueprint to the rest of the offshore waters of the Atlantic and Gulf.

Coastal & Marine

- **Atlantic coral and hardbottom** was added as a new coastal and marine indicator, replacing South Atlantic hardbottom and deep-sea coral. This new indicator expanded its spatial extent to all U.S. Atlantic waters within the SECAS geography, expanded its scope to consider human-created hardbottom habitat, and used slightly different data sources. We removed the NOAA deep-sea coral predictive habitat modeling data used in the previous version because newer National Oceanic and Atmospheric Administration (NOAA) data became available that depicted coral diversity, rather than the likelihood of occurrence for different taxa. This component split off into a different indicator called Atlantic deep-sea coral richness. We added several new datasets to Atlantic coral and hardbottom: newer NOAA hardbottom data from a recent deep-sea coral and hardbottom project, data on the cold-water coral mounds of the Blake Plateau provided by Dr. Derek Sowers, and NOAA shipwrecks and artificial reefs. This indicator continued to use The Nature Conservancy's (TNC) South Atlantic Bight Marine Assessment data, but only included the observed hardbottom polygons and not the high confidence, probable, potential, and possible hardbottom classes. It was too difficult to integrate these with the new NOAA hardbottom data, which used a different probabilistic approach. We continued to use the unpublished NOAA hardbottom data for shallower waters, but used the final logistic prediction layer classified into quantiles, rather than the thresholded logistic outputs, because that was more consistent with the new NOAA hardbottom data (based on guidance from Dr. Matt Poti). We also continued to use NOAA deep-sea coral point locations, but updated to a fresh download of that dataset to integrate any new points that had been added since the previous version. To accommodate these new and modified data sources, we changed our approach to scoring the indicator.
- **Atlantic deep-sea coral richness** was added as a new coastal and marine indicator to represent areas of high predicted deep-sea coral diversity. It is intended to complement Atlantic coral and hardbottom.
- **Atlantic marine birds** was added as a new coastal and marine indicator, replacing South Atlantic marine birds. This indicator expanded its spatial extent to all U.S. Atlantic waters within the SECAS geography. It used 10 distinct bins instead of a continuous ranking. We removed land to a separate class where the bird models overlapped with land areas identified in the National Land Cover Database. The new indicator also incorporated the latest source data from the Duke University's Marine-life Data and Analysis Team. The updated bird models allowed us to include three new species (bridled tern, parasitic jaeger, sooty tern), but also introduced an artifact for one species that we therefore had to remove (horned grebe). We

also resolved an error in the documentation that incorrectly referred to the brown pelican model as the Bermuda petrel model in previous iterations of the South Atlantic marine birds indicator. We mention this in the changelog to clarify that brown pelican was used across both versions, and did not replace the Bermuda petrel model as comparing the two versions of the documentation might suggest.

- **Atlantic marine mammals** was added as a new coastal and marine indicator, replacing South Atlantic marine mammals. This indicator expanded its spatial extent to all U.S. Atlantic waters within the SECAS geography. It used 10 distinct bins instead of a continuous ranking. The new indicator also incorporated the latest source data from the Duke Marine Lab, which used additional survey information and improved analysis methods and improved the spatial resolution from 10 km pixels to 5 km pixels. Using the latest models allowed us to include several new species, including seals, Atlantic spotted dolphin, Clymene dolphin, dwarf and pygmy sperm whales, pantropical spotted dolphin, and rough-toothed dolphin. The older beaked whale model was retired and replaced by three more taxonomically specific models for Cuvier's beaked whale, mesoplodont beaked whales, and unidentified beaked whales. Finally, we dropped the Sei whale model as the newer version introduced an artifact. The finer spatial scale of the new models allowed us to extend the indicator into nearshore waters, instead of clipping it to the marine subregions. As part of extending it nearshore, we removed land to a separate class where the mammal models overlapped with land areas identified in the National Land Cover Database.
- **Gulf coral and hardbottom** was added as a new coastal and marine indicator to capture the presence of coral and hardbottom in the Gulf. This is an analog to Atlantic coral and hardbottom.
- **Gulf deep-sea coral richness** was added as a new coastal and marine indicator to represent areas of high predicted deep-sea coral diversity in the Gulf. It is intended to complement Gulf coral and hardbottom.
- **Gulf marine mammals** was added as a new coastal and marine indicator to capture important areas in the Gulf for dolphins and whales. This is an analog to Atlantic marine mammals.
- **Gulf sea turtles** was added as a new coastal and marine indicator to capture important areas in the Gulf of America for sea turtles.
- **Marine highly migratory fish** was added as a new coastal and marine indicator to represent important foraging and spawning areas for highly migratory fish across the Atlantic and Gulf.

Caribbean

We developed many new indicators for the U.S. Caribbean in order to expand the Blueprint to Puerto Rico, the U.S. Virgin Islands, and the surrounding nearshore marine environment.

Terrestrial

- **Caribbean greenways and trails** was added as a new terrestrial indicator to capture the outdoor recreation and cultural resource value of greenways and trails in the U.S. Caribbean. This is an analog to the continental greenways and trails indicator, but uses some different source data and a lower maximum length threshold.
- **Caribbean habitat patch size (large islands)** was added as a new terrestrial indicator to capture natural habitat patches on large islands in the U.S. Caribbean. This is an analog to the continental intact habitat cores indicator, but uses different source data.
- **Caribbean habitat patch size (small islands)** was added as a new terrestrial indicator to capture natural habitat patches on small islands in the U.S. Caribbean. This is an analog to the continental intact habitat cores indicator, but uses different source data and a lower maximum size threshold.
- **Caribbean island habitat** was added as a new terrestrial indicator to represent the importance of island habitat for federally listed and other imperiled species in the U.S. Caribbean. This is an analog to the continental island habitat indicator, but uses some different source data and approaches.
- **Caribbean karst habitat** was added as a new terrestrial indicator to represent natural karst areas with limited human alteration in the U.S. Caribbean.
- **Caribbean landscape condition** was added as a new terrestrial indicator for the U.S. Caribbean to represent natural areas with limited human alteration while also considering the natural landcover of the surrounding landscape. It is complementary to the Caribbean habitat patch size indicators, which treat roads and developed areas as hard barriers rather than considering overall landscape permeability.
- **Caribbean low-urban historic landscapes** was added as a new terrestrial indicator to represent historic and cultural sites in the U.S. Caribbean that are surrounded by limited urban development. This is an analog to the continental South Atlantic low-urban historic landscapes indicator, but uses different source data.
- **Caribbean reforestation potential** was added as a new terrestrial indicator to represent opportunities to increase tree cover in the U.S. Caribbean based on current land uses and potential benefits for local drinking water supplies. It recognizes the important role that shade-grown coffee plays in the ecological and economic health of the region.
- **Caribbean urban park size** was added as a new terrestrial indicator to represent the cultural and recreational value of parks in urban areas in the U.S. Caribbean. This is an analog to the continental urban park size indicator, but uses some different source data and approaches.

Freshwater

- **Caribbean natural landcover in floodplains** was added as a new freshwater indicator to represent the amount of natural landcover in the estimated floodplain of rivers and streams within each catchment in the U.S. Caribbean. This is an analog to the continental natural landcover in floodplains indicator, but uses some different source data.
- **Caribbean network complexity** was added as a new freshwater indicator to capture the number of connected stream size classes in a river network between dams or waterfalls in the U.S. Caribbean. This is an analog to the continental network complexity indicator.
- **Caribbean permeable surface** was added as a new freshwater indicator to represent the average percent of non-impervious cover within each catchment in the U.S. Caribbean. This is an analog to the continental permeable surface indicator, but uses some different source data.

Coastal & Marine

- **Caribbean beach habitat** was added as a new coastal and marine indicator to represent the value of beaches that provide habitat for beach-nesting birds and sea turtles in the U.S. Caribbean.
- **Caribbean coastal shoreline condition** was added as a new coastal and marine indicator to assess shoreline alteration along the U.S. Caribbean coast. This is an analog to the continental shoreline condition indicator, but uses different source data and approaches.
- **Caribbean fish hotspots** was added as a new coastal and marine indicator to represent areas of high predicted fish density and diversity in the U.S. Caribbean where mangroves, seagrass, and coral occur in close proximity to one another.
- **Caribbean fish nursery habitat** was added as a new coastal and marine indicator to capture nursery and spawning habitat and other concentration areas for fish in the U.S. Caribbean.
- **Caribbean seagrass** was added as a new coastal and marine indicator to capture the presence and density of seagrass at various depths in the U.S. Caribbean. This is an analog to the continental seagrass indicator, but uses different source data and approaches.
- **Caribbean shallow hardbottom and coral** was added as a new coastal and marine indicator to represent hardbottom habitat and coral in the U.S. Caribbean and predict the ability of coral to survive the impacts of climate change. This is an analog to the Atlantic and Gulf coral and hardbottom indicators, but uses different source data and approaches.

Old Indicators Removed

Continental

Coastal & Marine

- **South Atlantic hardbottom and deep-sea coral** was removed as an indicator, replaced by Atlantic coral and hardbottom (see New Indicators Added section above).
- **South Atlantic marine birds** was removed as an indicator, replaced by Atlantic marine birds (see New Indicators Added section above).
- **South Atlantic marine mammals** was removed as an indicator, replaced by Atlantic marine mammals (see New Indicators Added section above).

Indicators Changed

Continental

Terrestrial

- **East Coastal Plain open pine birds** was updated to add a 0 class based on the extent of the source data. The indicator now more clearly shows the extent of the bird model from the East Gulf Coastal Plain Joint Venture. Since we corrected the oversight from the previous Blueprint where indicators were not used everywhere they occurred, this indicator was restored to its full extent and no longer clipped down to the subregions in which it was used in Blueprint 2022.
- **Equitable access to potential parks** was updated to add a 0 class to better show the analysis extent of the indicator. Because of the multiple interacting source datasets, the 0 class includes three types of areas: a) not assessed by ParkServe because it not in an urban area, b) assessed by ParkServe and found to be a low priority for a new park, or c) not in a socially vulnerable community.
- **Fire frequency** was updated to remove marine areas from the 0 class to show that marine areas are outside the scope of this terrestrial indicator.
- **Greenways & trails** was updated to a fresh download of the source data from OpenStreetMap to incorporate any new greenways, trails, sidewalks, and other paths that had been added since the previous version. We also used an improved method for distinguishing greenways and trails from sidewalks. Instead of using the name attribute to identify trails, we extracted all paths with a sidewalk tag to the sidewalk class and entirely removed all paths marked as private. All remaining bridleways, cycleways, footways, and paths were considered as greenways and trails and scored according to their connected length and natural condition.
- **Intact habitat cores** was updated to remove marine areas from the 0 class to better show that marine areas are outside the scope of this terrestrial indicator.

- **Mississippi Alluvial Valley forest birds – protection** was updated to add a 0 class based on the extent of the source data. The indicator now more clearly shows the extent of the bird model from the Lower Mississippi Valley Joint Venture. Since we corrected the oversight from the previous Blueprint where indicators were not used everywhere they occurred, this indicator was restored to its full extent and no longer clipped down to the subregions in which it was used in Blueprint 2022.
- **Mississippi Alluvial Valley forest birds - reforestation** was updated to add a 0 class based on the extent of the source data. The indicator now more clearly shows the extent of the bird model from the Lower Mississippi Valley Joint Venture. Since we corrected the oversight from the previous Blueprint where indicators were not used everywhere they occurred, this indicator was restored to its full extent and no longer clipped down to the subregions in which it was used in Blueprint 2022.
- **Playas** was updated to add a 0 class based on the boundary of the Playa Lakes Joint Venture, which provided the source data for this indicator.
- **South Atlantic amphibian & reptile areas** was updated to remove marine areas from the 0 class to show that marine areas are outside the scope of this terrestrial indicator. In addition, rather than clipping the Priority Amphibian and Reptile Conservation Areas (PARCAs) to the South Atlantic Landscape Conservation Cooperative (LCC) boundary, we kept the portions of all PARCAs that extended across the LCC border. Since we corrected the oversight from the previous Blueprint where indicators were not used everywhere they occurred, this indicator was restored to its full extent and no longer clipped down to the subregions in which it was used in Blueprint 2022.
- **South Atlantic forest birds** was restored to its full extent. Since we corrected the oversight from the previous Blueprint where indicators were not used everywhere they occurred, this indicator was no longer clipped down to the subregions in which it was used in Blueprint 2022.
- **South Atlantic low-urban historic landscapes** was restored to its full extent. Since we corrected the oversight from the previous Blueprint where indicators were not used everywhere they occurred, this indicator was no longer clipped down to the subregions in which it was used in Blueprint 2022.
- **Urban park size** was updated to add a 0 class based on the extent of the source data. We also made minor methods refinements to be more consistent with the approach used in the newly developed Caribbean version of the indicator. We buffered small parks that might otherwise be lost in the conversion from vector to raster. Compared to the previous version, this better represented small and/or linear features like Tallahassee-St. Marks Historic Railroad State Trail in the Big Bend of FL and Kingwood Greenway near Houston, TX. We also dissolved contiguous park units in an earlier step, which resulted in the inclusion of larger park complexes like Croatan National Forest in NC and Jean Lafitte National Wildlife Refuge in LA, whereas the previous version captured only discrete units of that park.
- **West Coastal Plain & Ouachitas forested wetland birds** was updated to add a 0 class based on the extent of the source data. The indicator now more clearly shows the extent of the bird

model from the Lower Mississippi Valley Joint Venture. Since we corrected the oversight from the previous Blueprint where indicators were not used everywhere they occurred, this indicator was restored to its full extent and no longer clipped down to the subregions in which it was used in Blueprint 2022.

- **West Coastal Plain & Ouachitas open pine birds** was updated to add a 0 class based on the extent of the West Gulf Coastal Plain/Ouachitas Bird Conservation Region. The indicator now more clearly shows the extent of the bird model from the Lower Mississippi Valley Joint Venture. Since we corrected the oversight from the previous Blueprint where indicators were not used everywhere they occurred, this indicator was restored to its full extent and no longer clipped down to the subregions in which it was used in Blueprint 2022.
- **West Gulf Coast mottled duck nesting** was updated to add a 0 class based on the extent of the source data. The indicator now more clearly shows the extent of the bird model from the Gulf Coast Joint Venture. Since we corrected the oversight from the previous Blueprint where indicators were not used everywhere they occurred, this indicator was restored to its full extent and no longer clipped down to the subregions in which it was used in Blueprint 2022.

Freshwater

- **Atlantic migratory fish habitat** was restored to its full extent. Since we corrected the oversight from the previous Blueprint where indicators were not used everywhere they occurred, this indicator was no longer clipped down to the subregions in which it was used in Blueprint 2022.
- **Gulf migratory fish connectivity** was updated to add a 0 class based on Gulf HUC6 watersheds, defining the analysis extent of the indicator. Since we corrected the oversight from the previous Blueprint where indicators were not used everywhere they occurred, this indicator was no longer clipped down to the subregions in which it was used in Blueprint 2022.
- **Imperiled aquatic species** was updated to add a 0 class representing all inland areas of the continental Southeast not identified as a floodplain. As the previous version of the indicator already had a 0 class depicting areas where 0 species had been observed, we reassigned that class a value of 1 and renumbered accordingly. We also updated to a fresh download of the source data from the Southeast Aquatic Resources Partnership (SARP), which had collected new data from many state National Heritage Programs since the last Blueprint update.
- **Natural landcover in floodplains** was updated to add a 0 class representing all inland areas of the continental Southeast not identified as a floodplain.
- **Network complexity** was updated to add a 0 class representing all inland areas of the continental Southeast not identified as a floodplain. We also updated to a fresh download of the source data from SARP. Because more source data was available for WV, we expanded this indicator to the entirety of WV whereas the previous version only covered the southern part of the state. We applied SARP's functional network lines to high resolution catchments instead of medium resolution catchments because the high resolution version aligned better with the linework. Finally, we attempted to fill gaps from the previous version of the indicator in places

where river networks contain loops or braided streams. Because SARP's network complexity analysis can only accommodate a one-way flow of water, they remove braided streams and loops from their calculations. As a result, some floodplain areas in the previous version of the indicator were missing a network complexity value. This year, we addressed this by identifying unscored floodplain areas and filling them in using the highest network complexity value from the surrounding pixels. This significantly improved the indicator in areas with complex or heavily modified hydrology, such as coastal Louisiana.

- **West Virginia imperiled aquatic species** was updated to add a 0 class representing all inland areas of the continental Southeast not identified as a floodplain. As the previous version of the indicator already had a 0 class depicting areas where 0 species had been observed, we reassigned that class a value of 1 and renumbered accordingly. In the source data, no areas within WV receive a value of 4 species observed, so in the previous version of the Blueprint, we skipped the number 4 in the legend, going directly from 3 species observed to 5 species observed. While this was accurate, the missing 4 value caused confusion for users and in online tools. As a result, in this update, we renumbered so that the highest value of the indicator is 4+ species observed to maintain the consecutive order of the classes.

Coastal & Marine

- **Atlantic estuarine fish habitat** was updated to include additional source data from the Atlantic Coast Fish Habitat Partnership that covered estuaries in southern FL. This additional input is analogous to the South Atlantic and Mid-Atlantic datasets already used in the previous version of the indicator, but had not been needed in previous iterations of this indicator when it only covered the South Atlantic. Since we corrected the oversight from the previous Blueprint where indicators were not used everywhere they occurred, this indicator was no longer clipped down to the subregions in which it was used in Blueprint 2022.
- **Coastal shoreline condition** was restored to its full extent. Since we corrected the oversight from the previous Blueprint where indicators were not used everywhere they occurred, this indicator was no longer clipped down to the subregions in which it was used in Blueprint 2022.
- **Estuarine coastal condition** was extended further offshore to include all estuarine waters shallower than 10 m in depth now that the Blueprint covers all U.S. marine waters within the SECAS geography. Previously, it was cut off at the boundary of the coastal Blueprint subregions. We also added a 0 class to better distinguish between estuarine areas that do not contain a National Coastal Condition Assessment sampling point and therefore are in unknown condition, and areas that are outside the extent of estuaries.
- **Islands** was renamed **island habitat** for consistency with the Caribbean version of the indicator. Since we corrected the oversight from the previous Blueprint where indicators were not used everywhere they occurred, this indicator was no longer clipped down to the subregions in which it was used in Blueprint 2022.

- **Resilient coastal sites** was restored to its full extent. Since we corrected the oversight from the previous Blueprint where indicators were not used everywhere they occurred, this indicator was no longer clipped down to the subregions in which it was used in Blueprint 2022.
- **Seagrasses** was renamed **seagrass** for consistency with the Caribbean version of the indicator. We also updated it to a fresh download of the source data from the NOAA Marine Cadastre to incorporate any seagrass areas that had been added since the previous version. It was extended further offshore to include all source data within the continental SECAS geography, now that the Blueprint covers all U.S. marine waters in the Gulf and Atlantic. Previously, it was cut off at the boundary of the coastal Blueprint subregions.
- **South Atlantic beach birds** was updated to improve the spatial resolution by recalculating the indicator from the original bird data for Wilson’s plover and American oystercatcher, and intermediate 90 m outputs for least tern and piping plover. The previous version of this indicator was resampled from a 200 m layer dating back to South Atlantic Blueprint 2.0. We also clipped out areas with poor data coverage in Virginia and the South Atlantic portion of the Gulf coast. Since we corrected the oversight from the previous Blueprint where indicators were not used everywhere they occurred, this indicator was no longer clipped down to the subregions in which it was used in Blueprint 2022.
- **South Atlantic maritime forest** was updated to add a 0 class to represent the extent of the indicator. We modified that extent from the previous version by applying it to the Blueprint subregions that roughly correspond to the South Atlantic LCC geography, instead of the exact South Atlantic LCC boundary. This extended the indicator to parts of Virginia that were not covered in the previous version. The new extent is more ecologically than administratively based, while staying broadly within the portion of the SECAS region where it has received expert review. Since we corrected the oversight from the previous Blueprint where indicators were not used everywhere they occurred, this indicator was no longer clipped down to the subregions in which it was used in Blueprint 2022.
- **Stable coastal wetlands** was updated to add a 0 class based on the extent of the source data. We also repurposed the 1 class to encompass other coastal wetlands that were not identified as stable, to better depict all assessed wetlands. The top stable coastal wetlands class was renumbered accordingly to a value of 2. Since we corrected the oversight from the previous Blueprint where indicators were not used everywhere they occurred, this indicator was no longer clipped down to the subregions in which it was used in Blueprint 2022.

Changes to Connectivity

We continued to use Linkage Mapper for our connectivity analysis in Southeast Blueprint 2023.

- We upgraded to use the latest version of Linkage Mapper, version 3.1.0. The previous version of the Blueprint used Linkage Mapper 3.0.
- Hubs and corridors were expanded to new areas of the Blueprint—U.S. Gulf and Atlantic offshore waters, Puerto Rico and the U.S. Virgin Islands, and the surrounding nearshore marine environment of the U.S. Caribbean.
- Southeast Blueprint 2023 priorities were used to define the hubs and resistance rasters across the inland continental, marine continental, and Caribbean connectivity analyses.

Continental

Inland Hubs & Corridors

We made several changes to the inland continental connectivity analysis:

- We made a minor refinement to our wildlife road crossing methods. We lowered the resistance value of the roads surrounding the road crossings from 500 to 150. This allowed us to use a 4-bit raster for the resistance raster rather than an 8-bit, which reduced file size and potentially sped up the analysis.
- We used the same size of hexagons in the inland connectivity analysis as in the previous version, but their placement shifted slightly so that the hexagons do not align across 2022 and 2023. This did not affect the final results.

Marine/Estuarine Hubs & Corridors

We made several changes to the marine/estuarine continental hubs and corridors analysis:

- Marine hubs and corridors were expanded to the full Gulf and Atlantic offshore marine environment. The previous Blueprint's marine connectivity analysis was restricted to a portion of the Atlantic marine environment.
- The previous marine mammal movement zones became marine mammal and turtle movement zones and used new and updated source data. The Atlantic marine mammal data was updated to the latest models from the Duke Marine Lab used in the new Atlantic marine mammals indicator. The three Atlantic species used in 2022 were North Atlantic right whale, sperm whale, and humpback whale. The 2023 movement zones continued to use right whale and humpback whale, but replaced sperm whale with pilot whale, keeping the same abundance thresholds. To expand this layer to the Gulf, we also incorporated four species models (Rice's whale, pilot whale, Kemp's ridley sea turtle, and loggerhead sea turtle) from the Gulf of Mexico Marine Assessment Program for Protected Species (GoMMAPPS) used in the new Gulf marine mammals and Gulf sea turtles indicators.

Caribbean

- Hubs and corridors were expanded to the U.S. Caribbean. These were run in a separate Caribbean connectivity analysis. Because this analysis was new for 2023, there is no previous version to compare it to. However, it is important to note that, compared to the continental hubs and corridors approach, the Caribbean connectivity analysis used much simpler methods. It ran all hubs and all corridors together in one seamless analysis, rather than running marine and inland areas separately. It did not use all nearshore estuaries as hubs, and did not use marine mammal movement zones to constrain the resistance raster. We intend to build on these simpler methods as a model for streamlining the continental connectivity analysis in future Blueprint updates. The Caribbean hubs also used a smaller size threshold to reflect the smaller size of the geography compared to the continental Southeast (200 ha/~500 ac instead of 2,000 ha/~5,000 ac).

Changes to the Zonation Results

Improvements to the Overall Zonation Modeling Approach

Blueprint 2023 continued to use Zonation to prioritize within subregions, with several improvements.

- Because Blueprint 2023 included new areas of the offshore marine environment and U.S. Caribbean, we modified the subregions used to run Zonation. We removed the previous Marine Shelf and Extension and Marine Gulf Stream subregions, which applied only to the portion of the Atlantic marine environment covered by Base Blueprint 2022. They were replaced by an Atlantic Marine subregion that covered the entire U.S. Atlantic marine environment within the SECAS geography. We added a new Gulf of America subregion that covered the entire U.S. Gulf marine environment within the SECAS geography. We also added a new Caribbean subregion that covered Puerto Rico, the U.S. Virgin Islands, and the surrounding nearshore U.S. Caribbean marine environment.

Continental

Inland

In the inland continental area, Blueprint 2023 continued to use Zonation 4, with the same settings, to prioritize within the same subregions. However, we adjusted the weighing rules as follows:

- The goal of the Zonation weighting rules remained the same (to ensure that spatially limited indicators were not overprioritized and that significantly outdated indicator data had less of an influence). However, we had to add several new exceptions to the standard spatially limited indicator weighting rule to accomplish the first objective for Blueprint 2023.

- We modified the use of the spatially limited indicators rule for imperiled aquatic species in the High Plains and Tablelands subregion so that it was applied to ≥ 2 values instead of ≥ 1 . This increased the weight for the indicator in this subregion.
- The natural landcover in floodplains indicator in the Ouachita subregion fell on the other side of the spatially limited indicator weighting rule in 2023, which lowered its weight significantly compared to 2022. To maintain better consistency within the 2022 priorities in this subregion, we applied the spatially limited indicators weighting rule for indicators where the proportion of ≥ 1 values is ≤ 0.1 , instead of the one where the proportion of ≥ 1 values is 0.5 to 0.1.
- We resolved an error in the documentation that failed to capture a weighting rule that was also used in 2022. The Interior Southeast grasslands indicator needed an adjusted weight in the North Appalachians, South Appalachians, Piedmont, and Ozarks and Plains subregions. In these four subregions, where the top indicator class (known grasslands) covered less than 0.03% of the area, lower indicator values were having too strong of an impact on the subregion's priorities. To address this, we multiplied the weight of Interior Southeast Grasslands by 0.5 in those subregions. We mention this in the changelog to clarify that this is not a new rule for 2023, as comparing the two versions of the documentation might suggest.
- Because of the changes made to inland continental indicators in 2023, many of the indicator weights themselves changed. However, we did not document those here because the weights are highly variable from year to year. The changelog only captures changes to the rules that are used to calculate those weights.

Marine

The approach to running Zonation in the offshore marine area changed significantly, with new subregions, a new version of Zonation, and modified goals and rules for weighting indicators.

- We ran Zonation within a new combined Atlantic Marine and Gulf of America subregion.
- In the new marine and Caribbean subregions, we used a new and improved version of Zonation: v5. Zonation 5 offers different settings and rules compared to Zonation 4, but we used the setting most consistent with the core-area Zonation algorithm used in the inland subregions. The most notable difference is that edge removal is no longer available in Zonation 5, which affects the aggregation of priorities in the final Blueprint. Zonation 5 also no longer requires specifying a warp factor. Zonation 5 is much faster and less computationally demanding than Zonation 4, which makes it possible to run across larger areas. We hope to use Zonation 5 for the entire Blueprint in the next update, but wanted to minimize changes in the inland subregions between version 2022 and 2023 because workshops for Blueprint 2023 focused on the Caribbean and marine areas.
- In the marine subregions, we added a new indicator weighting goal: balancing priorities across Gulf and Atlantic areas of the offshore U.S. marine environment. To achieve this, a new weighting rule was introduced that applies to indicators that cover only the Atlantic or Gulf.

We only weighted bird, mammal, and turtle indicators when the proportion of ≥ 1 values was less than 0.3. We also increased the weight of two indicators that were getting significantly underrepresented in the Blueprint priorities (Atlantic coral and hardbottom, Gulf deep-sea coral richness) from 0.3 to 0.75.

- Because of the changes made to marine continental indicators in 2023, many of the indicator weights themselves changed. However, we did not document those here because the weights are highly variable from year to year. The changelog only captures changes to the rules that are used to calculate those weights.

Caribbean

Because Base Blueprint 2022 did not cover any part of the U.S. Caribbean, there is no previous version in this area to compare Blueprint 2023 to. However, it is important to note that we ran the Caribbean Zonation analysis within a new Caribbean subregion that covered both inland and offshore areas, using Zonation 5 and the same settings as the marine continental analysis. In this subregion, we had only one goal for indicator weighting: to ensure spatially limited indicators were not overprioritized. We were able to use only the standard spatially limited indicators weighting rule: for indicators where the proportion of ≥ 1 values was 0.5 and higher, we set their weight to 1.0. No special exceptions were necessary.

Southeast Blueprint 2022

Version 2022 of the Southeast Blueprint was released in October 2022. This version used a fundamentally different approach. Instead of stitching together so many different subregional plans, we took major strides toward SECAS-wide consistency by using the same methods and data across all 15 Southeast states. This portion of the Southeast Blueprint is referred to as the "Base Blueprint". The Base Blueprint expanded and improved the approach used to develop the South Atlantic Blueprint. More detail on the differences between South Atlantic Blueprint 2021 and Base Blueprint 2022 is available in the [South Atlantic/Base Blueprint Changelog](#).

At the highest level, significant improvements over the previous versions of the Southeast Blueprint include:

- Improved priorities across the 15-state Base Blueprint area
- A connectivity analysis covering all 15 SECAS states
- Improved priorities for the Florida marine environment

The 2022 Blueprint update also used different priority classes and wording. We switched from “conservation value” to “priorities for a connected network of lands and waters” based on feedback from SECAS staff, science communication professionals, Southeast Blueprint users, and [workshop attendees](#). This new wording better aligns with the SECAS vision statement.

In addition, because we no longer had to stitch together so many different input plans in 2022, we increased the number of priority classes to match those used in the Base Blueprint. Instead of two classes, highest and high conservation value, the 2022 Blueprint has four to better stratify the landscape: highest priority, high priority, medium priority, and priority connections.

Changes to Subregional Inputs

With the shift to consistent methods and data across most of the SECAS geography, the number of subregional plans stitched together in the Southeast Blueprint decreased from 10 to 3. Only the Caribbean Landscape Conservation Design remained unchanged from Southeast Blueprint 2021. The following subregional inputs did change in Southeast Blueprint 2022:

- South Atlantic Blueprint (replaced by Base Blueprint 2022)
- Florida Marine Blueprint (updated to new version)

Changes & Updates to Inputs

Base Blueprint 2022

The new input plan that uses consistent methods and indicators across all 15 Southeast states is referred to as the “Base Blueprint” because it forms the base of the Southeast Blueprint. This plan replaced the South Atlantic Blueprint, Appalachian NatureScape Design, Florida Blueprint, Gulf Hypoxia Blueprint, Middle Southeast Blueprint, North Atlantic Nature’s Network, Oklahoma CHAT, and Texas CHAT. The Base Blueprint applies a modified version of the 2021 South Atlantic Blueprint approach. For more information on the changes made between South Atlantic Blueprint 2021 and Base Blueprint 2022, [see the South Atlantic/Base Blueprint changelog](#).

Florida Marine Blueprint

A new version of the Florida Marine Blueprint (version 2.0) replaced Florida Marine Blueprint 1.0 within the Florida marine environment.

Integration of Overlapping Southeast Blueprint Inputs

Changes to Inputs Used in Specific Overlap Zones

The integration rules and overlap zones used in previous versions of the Southeast Blueprint were no longer needed in 2022. The only overlap between input plans occurred between the Base Blueprint and Florida Marine Blueprint in the nearshore waters around Florida. The Base Blueprint was used anywhere it occurred, and the Florida Marine Blueprint and Caribbean Landscape Conservation Design were used only outside of the Base Blueprint area.

Changes to Input Areas Layer

The input areas layer was updated to reflect the extents of the updated inputs to the Blueprint.

Changes to Connectivity Layer

The Base Blueprint provided a consistent connectivity layer that covered all 15 SECAS states. This layer is referred to as “hubs and corridors” rather than the “hubs and connectors” wording used in previous versions of the Southeast Blueprint, because it is an extension of the previous South Atlantic Blueprint connectivity methodology.

Southeast Blueprint 2021

Version 2021 of the Southeast Blueprint was released in November 2021. At the highest level, significant improvements over the previous versions include:

- Correcting a spatial rebalancing error in the Middle Southeast subregion by using the full Middle Southeast Blueprint 2020 as the input to Southeast Blueprint 2021
- Major improvements to the South Atlantic Blueprint, including:
 - Updated and improved indicators
 - Better incorporation of equity, deep-sea coral, fire, and important grassland areas
 - Multiple connectivity-related improvements

Changes to Subregional Inputs

Florida Blueprint 1.3, Florida Marine Blueprint 1.0, Gulf Hypoxia Precision Conservation Blueprint v1.5, the Appalachian NatureScape Design and The Nature Conservancy's Resilient and Connected Landscapes "Prioritized Network" layer, Nature's Network, the Caribbean Landscape Conservation Design, Oklahoma CHAT, and Texas CHAT remain unchanged from Southeast Blueprint 2020. The following subregional inputs did change in Southeast Blueprint 2021.

Changes & Updates to Inputs

South Atlantic Blueprint 2021

An updated version of the South Atlantic Blueprint was used. This version is named South Atlantic Blueprint 2021. South Atlantic Blueprint uses improved data for many indicators, an improved connectivity approach, more consistent indicator weighting, and a finer spatial resolution in the marine environment.

Middle Southeast Blueprint 2020

The previous Southeast Blueprint used a version of Middle Southeast Blueprint 2020 that was spatially rebalanced within the smaller extent where the Middle Southeast Blueprint was used as an input to the Southeast Blueprint. Unfortunately, an error was made in the spatial rebalancing process. To correct that error in Southeast Blueprint 2021, we used the full (non-rebalanced) [Middle Southeast Blueprint 2020](#) as the input to the Middle Southeast region.

Integration of Overlapping Southeast Blueprint Inputs

Changes to Inputs Used in Specific Overlap Zones

While the integration rules themselves remained the same as those used in Blueprint 2020, the application of those rules resulted in different inputs being used in specific overlap zones.

Plans used in the following overlap zones changed this year due to changes in the extent of the South Atlantic Blueprint:

- A portion of the nearshore environment on the Gulf coast of Florida was included in the 2021 South Atlantic Blueprint for the first time, so South Atlantic Blueprint priorities were used in that area instead of Florida Marine Blueprint priorities.
- A portion of northern Florida near Jacksonville was not included in the South Atlantic Blueprint this year. As a result, this area is no longer an overlap zone, so Florida Blueprint priorities were used alone; the previous version of the Southeast Blueprint used both the Florida Blueprint and South Atlantic Blueprint priorities.

For comparison, plans used in the following overlap zones remained the same this year:

- The zone in MO where the Middle Southeast Blueprint and Gulf Hypoxia Blueprint overlap: both plans continued to be used.
- The zone in VA where the South Atlantic Blueprint and Nature's Network overlap: both plans continued to be used.
- The zone in the Big Bend of FL where the South Atlantic Blueprint and Florida Blueprint overlap: both plans continued to be used.
- The zone in the Big Bend of FL where the South Atlantic Blueprint, Florida Blueprint, and Middle Southeast Blueprint overlap: all three plans continued to be used.

Changes to Input Areas Layer

The input areas layer was updated to reflect the extents of the updated inputs to the Blueprint, as well as the inputs used in the overlap zones.

Changes to Connectivity Layer

The connectivity layer incorporated updated hubs and connectors for the South Atlantic subregion.

Southeast Blueprint 2020

Version 2020 of the Southeast Blueprint was released at the October 2020 SEAFWA virtual annual meeting. At the highest level, significant improvements over the previous versions include:

- Major improvements to the inland area covered by the South Atlantic Blueprint, including finer resolution and a more connected network of priorities.
- Improvements to the area covered by the Middle Southeast Blueprint, including updated data and an approach to overall prioritization that will make comparisons across the Southeast more consistent.
- Better integration in areas where the South Atlantic Blueprint overlaps with the Florida Blueprint and with Nature's Network.

Changes to Subregional Inputs

Florida Blueprint 1.3, Florida Marine Blueprint 1.0, Gulf Hypoxia Precision Conservation Blueprint v1.5, the Appalachian NatureScape Design and The Nature Conservancy's Resilient and Connected Landscapes "Prioritized Network" layer, Nature's Network, the Caribbean Landscape Conservation Design, Oklahoma CHAT, and Texas CHAT remain unchanged from Southeast Blueprint 4.0. The following subregional inputs did change in Southeast Blueprint 2020.

Changes & Updates to Inputs

South Atlantic Blueprint 2020

An updated version of the South Atlantic Blueprint was used. This version is named South Atlantic Blueprint 2020. South Atlantic Blueprint uses a finer spatial resolution, improved modeling approach, and improved data for many indicators.

Middle Southeast Blueprint 2020

An updated version of the Middle Southeast Blueprint was used. Note: This input has previously been referred to as the Middle South Blueprint and Gulf Coastal Plains & Ozarks (GCPO) Blueprint; the name has now changed to Middle Southeast. The 2020 Middle Southeast Blueprint includes minor improvements to the underlying species index, conservation hubs (partnership focal areas) layer, and risk index; it also prioritizes the species index, habitat index, and conservation hubs more highly in the final Blueprint. The Middle Southeast Blueprint was also spatially rebalanced this year within the smaller extent where the Middle Southeast Blueprint is used as an input.

Integration of Overlapping Southeast Blueprint Inputs

Changes to Inputs Used in Specific Overlap Zones

While the integration rules themselves remained the same as those used in Blueprint 4.0, the application of those rules resulted in different inputs being used in specific overlap zones.

Plans used in the following overlap zones changed this year:

- The zone in KY, TN, and AL where the Middle Southeast Blueprint and Appalachian NatureScape Design overlap: both plans were used this year, while only the Middle Southeast Blueprint was used in the previous version.
- The zone in AL, GA, NC, and SC where the South Atlantic Blueprint and Appalachian NatureScape Design overlap: only the App was used this year, while only the South Atlantic was used last year. Note: this overlap zone is much larger this year because the South Atlantic Blueprint was expanded to cover a larger buffer area.
- The zone in GA and AL where the Middle Southeast Blueprint and South Atlantic Blueprint overlap: both plans were used this year, while only the South Atlantic Blueprint was used in the previous version.
- The zone in the westernmost panhandle of Florida where the Florida Blueprint and Middle Southeast Blueprint overlap: both plans were used this year, while only the Florida Blueprint was used last year.

For comparison, plans used in the following overlap zones remained the same this year:

- The zone in MO where the Middle Southeast Blueprint and Gulf Hypoxia Blueprint overlap: both plans continued to be used.
- The zone in VA where the South Atlantic Blueprint and Nature's Network overlap: both plans continued to be used.
- The zone in the Big Bend of FL where the South Atlantic Blueprint and Florida Blueprint overlap: both plans continued to be used.
- The zone in the Big Bend of FL where the South Atlantic Blueprint, Florida Blueprint, and Middle Southeast Blueprint overlap: all three plans continued to be used.

Changes to Input Areas Layer

The input areas layer was updated to reflect the extents of the updated inputs to the Blueprint, as well as the inputs used in the overlap zones.

Changes to Connectivity Layer

The connectivity layer incorporated updated hubs and connectors for the South Atlantic subregion.

Southeast Blueprint v4.0

Version 4.0 of the Southeast Blueprint was released at the October 2019 SEAFWA annual meeting. At the highest level, significant improvements over the previous versions include: corrected overprioritization in Texas, Oklahoma, and the mountains of West Virginia and Virginia; improved priorities in the Lower Mississippi Valley, Louisiana marshes, and the Southern Appalachians; updated inputs from Florida and the Middle South subregion; expanded marine coverage to include state and federal waters around Florida; and expanded hubs and corridors that now cover all of Florida.

Changes to Subregional Inputs

South Atlantic Blueprint 2.2, Nature's Network, the Caribbean Landscape Conservation Design, and the Oklahoma CHAT remain unchanged from Southeast Blueprint 3.0. The following subregional inputs did change in Blueprint 4.0.

New Inputs Added

Florida Marine Blueprint

A new marine Blueprint for Florida was added as an input. This is the first marine Blueprint for this area, so it is named Florida Marine Blueprint 1.0.

Inputs Removed

The Gulf Coast Prairie (GCP) Blueprint 1.0

The Gulf Coast Prairie Blueprint is no longer used in Version 4.0 of the Southeast Blueprint. With the expansion of the Middle South (former GCPO) Blueprint to cover all of Louisiana, the GCP subregion is entirely captured by other Blueprint inputs. There is currently no process identified to update the GCP Blueprint long-term. The integration rules chosen by the expert teams that reviewed the Texas/Oklahoma and Gulf of America overlap areas chose to exclude the GCP Blueprint and use only the Crucial Habitat Assessment Tool (CHAT) and Middle South Blueprint.

Changes & Updates to Inputs

Florida Blueprint

An updated version of the Florida Blueprint was used. This version is named Florida Blueprint 1.3.

The Appalachian NatureScape Design

We continue to use data from the Phase II Appalachian NatureScape Design. Because NatureScape does not cover enough of the landscape to be consistent with the other inputs, we augment it with

another prioritization. In Blueprint 4.0, we augmented the NatureScape design with The Nature Conservancy's Resilient and Connected Landscapes "Prioritized Network" layer instead of the NatureScape Phase I Species Richness layer that was used in 3.0. This newly available layer does a better job identifying biodiversity hotspots that will be resilient to climate change, and, unlike the species richness layer, all the underlying information is also publicly available.

Middle South Blueprint (previously called the Gulf Coastal Plains & Ozarks, or GCPO, Blueprint)

The new Middle South Blueprint is an updated, expanded, and renamed version of the Gulf Coastal Plains and Ozarks (GCPO) Blueprint. It now covers the entire state of Louisiana and has significantly improved priorities in the Lower Mississippi Valley and the marshes of Louisiana.

Gulf Hypoxia Precision Conservation Blueprint v1.5

We continue to use data from the Gulf Hypoxia Precision Conservation Blueprint v1.5 Sum - Conservation and Watershed Interests (2016) layer. In version 3.0 of the Southeast Blueprint, scores of 5-7 were defined as "high", accidentally omitting scores of 8. This error has been corrected in Version 4.0 of the Southeast Blueprint, and high is now defined as scores of 5-8. Medium remains unchanged and is still defined as a score of 4.

The Crucial Habitat Assessment Tool for Texas

We continue to use data from the CHAT to cover both Texas and Oklahoma. While the Oklahoma CHAT input to Southeast Blueprint 4.0 remains unchanged, we changed our approach to spatially rebalancing the Texas CHAT input to the Southeast Blueprint to address issues with overprioritization. In Version 3.0 of the Southeast Blueprint, scores of 1 and 2 on the combined CHAT rank were defined as high, and scores of 3 were defined as medium. This approach continues to work well in Oklahoma, but in Texas, this prioritized too much of the landscape as high (more than the 30% intended). In the Version 4.0 Texas CHAT input, "high" includes CHAT rank 1 and only a subset of CHAT rank 2 with high terrestrial and aquatic scores (aquatic 2 & terrestrial 2, aquatic 2 & terrestrial 3, aquatic 3 & terrestrial 2). "Medium" is defined as the remainder of CHAT rank 2 with lower terrestrial and aquatic scores. CHAT rank 3 is no longer prioritized in the Texas CHAT input to Southeast Blueprint 4.0.

Integration of Overlapping Southeast Blueprint Inputs

Changes to Integration Rules

We use integration rules to define how Blueprint priorities are determined in areas where multiple subregional inputs overlap. Several changes were made to these integration rules in Southeast Blueprint 4.0.

In Southeast Blueprint 3.0, if an area was a priority in any of the inputs, it was included in the final Blueprint. If two inputs disagreed on whether an area should be medium or high value, we used the score from the most mature plan (the input that has been in development the longest) to favor the most well-established plans. This rule resulted in overlap areas that were overprioritized, with much more than 50% of the landscape included in the final Southeast Blueprint. We expect some overlap areas to slightly exceed the 50% target for subregional inputs due to the conservation importance of some of these areas. In some cases, however, more than 90% of the overlap area was prioritized.

To reduce overprioritization and confusion about what drives the Blueprint priorities, we instead only included more than one input in an overlap zone if known Blueprint uses had occurred using the input in that overlap zone. If no known Blueprint uses had occurred for any of the inputs, we included only the most well-established input. Teams of local experts reviewed the spatial results of the integration rules and approved the final approach. These improved methods massively reduced the amount of overprioritization. For example, in the West Virginia and northern Virginia overlap zone, the percentage of the landscape prioritized in the Blueprint decreased from 80% to 55%.

Changes to Inputs Used in Specific Overlap Zones

The new broad integration rules resulted in different plans being used in specific overlap zones. The treatment of overlap zones in Southern Virginia and Southern Missouri are the only overlap zones that did not change in Blueprint 4.0. Southern Virginia still uses both South Atlantic and Nature's Network inputs where those plans overlap.

Similarly, Southern Missouri still uses both the Middle South and Gulf Hypoxia where both plans overlap. However, while continuing to use the Gulf Hypoxia Blueprint across the full state of Missouri in Southeast Blueprint 4.0 does not reflect a change in methodology, this does reflect a change in approach. In Southeast Blueprint 3.0, using the Gulf Hypoxia Blueprint statewide was an error; it was not intended to be used in the overlap zone with the Gulf Coastal Plains and Ozarks Blueprint, but only to fill the gap in northern Missouri not covered by any other inputs. In Blueprint 4.0, the Gulf Hypoxia Blueprint is used statewide because there are use cases for that plan in the overlap zone with the Middle South Blueprint.

The approach to the following overlap zones did change in Southeast Blueprint 4.0:

- The panhandle of Florida now uses only the South Atlantic and Florida Blueprints where those plans overlap. The Middle South Blueprint is no longer used where it overlaps with the Florida Blueprint.
- Eastern Alabama/Southwest Georgia now only uses the South Atlantic Blueprint. The Middle South Blueprint is no longer used where it overlaps with the South Atlantic Blueprint.
- The Louisiana coast now only uses the Middle South Blueprint. Now that the Middle South Blueprint has expanded to cover the entire state of Louisiana, the Gulf Coast Prairie Blueprint is no longer used where it overlaps with the Middle South Blueprint.

- West Virginia and Northern Virginia now only uses the Nature’s Network Blueprint. The Appalachian NatureScape design is no longer used where it overlaps with Nature’s Network.
- Eastern Texas now only uses the Texas CHAT. The Middle South and Gulf Coast Prairie Blueprints are no longer used where they overlap with the Texas CHAT.
- Eastern Oklahoma now only uses the Oklahoma CHAT. The Middle South and Gulf Coast Prairie Blueprints are no longer used where they overlap with the Oklahoma CHAT.

Other Miscellaneous Changes

Because we no longer use a universal integration rule for the CHAT, and instead use different integration rules for Texas and Oklahoma, we no longer include the CHAT for other states when merging together the Southeast Blueprint before it is clipped down to the SECAS geography.

Changes to Connectivity Layer

The connectivity layer for Blueprint 4.0 incorporated new hubs and connectors data from the improved Florida Blueprint 1.3. Because “connectors/connections” was the most commonly used word, rather than “corridors”, this layer was renamed “hubs and connectors” rather than “hubs and corridors”.

Changes to Input Areas Layer

The input areas layer was updated to reflect the extents of the updated inputs to the Blueprint, as well as the improved approaches for dealing with multiple inputs in overlap zones. Since overprioritization is no longer a significant problem in these overlap zones, this layer was renamed simply “input areas” instead of “input and overlap areas”.

We added some new fields in the attribute table to help communicate how inputs were used in creating the Blueprint 4.0. We now have a field named “InputsUsedIn_4_0”, which lists the inputs that were actually used in each area to create Southeast Blueprint 4.0. This is in addition to the “InputExtents” field, which shows the full extent of each input area, but does not reflect whether the input was used in the Southeast Blueprint in that area. In Blueprint 3.0, we had a different field called “SEBlueprintInputAll,” which served both purposes, because the old integration rules used each input wherever it occurred.

In addition, we changed the default display for the input areas layer to show where each subregional input is used in Blueprint 4.0, rather than the full extent of each subregional input. The attributes of this layer can still be used to show the full extent of each subregional input, but that is no longer the primary function of this layer since issues with overprioritization in overlap zones have now been mostly resolved. The input areas layer is now primarily intended to serve as a resource for

determining which of the subregional Blueprint inputs covers a particular area of interest, providing a roadmap for where to look for more detailed information.

We've done our best with the following section, but keep in mind it does not represent an exhaustive list of every single change in every past version of the Blueprint. It contains all the changes we documented and were able to recall. The list is more comprehensive with more recent versions as Blueprint documentation improved over time and staff memories were fresher. In other words, we're almost certainly missing a few things, but this should be pretty close! It's always easier to maintain a thorough record moving forward than it is to try to reconstruct the past.

Southeast Blueprint v3.0

Version 3.0 of the Southeast Blueprint was first introduced at the October 2018 SEAFWA annual meeting, and officially released in February 2019. At the highest level, significant improvements over the previous versions include: full coverage of Texas, integrated threat layers covering the full Southeast, and the addition of a hubs and corridors layer covering part of the region.

Changes to Subregional Inputs

New Inputs Added

The Crucial Habitat Assessment Tool for Texas

The CHAT was expanded to Texas and incorporated into Southeast Blueprint 3.0. This filled in a gap in priorities in Western Texas and ensured complete coverage of all SEAFWA states.

Changes to Complementary Data Layers

Several new complementary data layers were developed and added to the Blueprint data gallery:

- A new layer depicting Blueprint input and overlap areas
- An integrated connectivity layer that combined hubs and corridors for the South Atlantic and Appalachian subregions
- Threats and land use change layers, including solar energy suitability, urban growth, and sea-level rise inundation

Other Miscellaneous Changes

We updated and improved the process of mosaicing together the various Blueprint inputs for better spatial accuracy. This included improving the methods we used to convert vector inputs to rasters and being more deliberate with methods for reprojecting and resampling raster inputs. This results in the Southeast Blueprint more closely aligned with its inputs.

Southeast Blueprint v2.0

Southeast Blueprint 2.0 was presented in October during the 2017 Southeastern Association of Fish and Wildlife Agencies' annual meeting, and officially released in November 2017. At the highest level, significant improvements over version 1.0 include improved consistency across LCC boundaries, improved consistency in climate change response, and improved integration beyond the Southeast.

Changes to Subregional Inputs

New Inputs

The Crucial Habitat Assessment Tool

Southeast Blueprint 2.0 brought in priorities from the Western Association of Fish and Wildlife Agencies' Crucial Habitat Assessment Tool (CHAT) to integrate beyond the Southeast (note: CHAT data for the state of Texas was not yet available for Blueprint 2.0).

Changes & Updates to Inputs

The Appalachian NatureScape Design

Southeast Blueprint 1.0 used the species richness layer from Phase I of NatureScape as the input for the Appalachian subregion. Blueprint 2.0 used the updated Phase II NatureScape design. Because the Phase II design did not cover enough of the landscape to be consistent with the other inputs, we augmented it with the same Phase I species richness layer used in 1.0. The Phase II NatureScape design is a connectivity-driven plan, so this added an important connectivity component to the Southeast Blueprint.

The South Atlantic Blueprint

An updated version of the South Atlantic Blueprint was used. This version was named South Atlantic Blueprint 2.2. This version of the South Atlantic Blueprint incorporated improvements to several indicators and minor refinements to the modeling approach.

Southeast Blueprint v1.0

Southeast Blueprint 1.0 was presented in October during the 2016 Southeastern Association of Fish and Wildlife Agencies' annual meeting in Baton Rouge, Louisiana, and officially released in December 2016. This plan provided the first ever integration of spatial plans developed through the South Atlantic, Appalachian, Gulf Coastal Plains and Ozarks, Gulf Coast Prairie, North Atlantic, Peninsular Florida, and Caribbean LCCs. It also included the spatial plan from the Mississippi River Basin/Gulf Hypoxia Initiative spearheaded by seven LCCs (Appalachian LCC, Eastern Tallgrass Prairie and Big Rivers LCC, Great Plains LCC, Gulf Coast Prairie LCC, Gulf Coastal Plains and Ozarks LCC, Plains and Prairie Potholes LCC, and the Upper Midwest and Great Lakes LCC).